Real-Time Texture Recognition Using Machine and Deep Learning: A Comparative Study from Hybrid Models to Vision Transformers

Submitted by

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ABSTRACT

Texture recognition plays an important role in camera-based industrial inspection, especially in scenarios such as semiconductor and surface-quality monitoring, where subtle appearance differences must be identified accurately and efficiently. However, practical texture classification remains difficult because texture cues are often fine-grained, repetitive, and sensitive to limited training data, domain variation, and image noise. In view of these challenges, this final year project investigates the development of real-time texture recognition methods from conventional convolutional neural networks (CNNs) and hybrid machine-learning/deep-learning pipelines to recent vision Transformer models. The study focuses on classification accuracy, F1-score, confusion-matrix analysis, real-time feasibility, and robustness to noise.

This thesis forms part of a joint project on real-time texture recognition for industrial inspection. The present work concentrates on the progression from deep feature extraction and hybrid classification strategies to hierarchical Transformer architectures. The investigation is conducted in two stages. In the first stage, experiments on the Describable Textures Dataset (DTD) are used to examine the small-sample problem. End-to-end training with ResNet-50 showed clear overfitting, with training accuracy reaching 82% while validation accuracy remained around 65%. MobileNetV2 combined with Mixup reduced overfitting but introduced underfitting, with performance converging at approximately 64%. To address this limitation, a hybrid approach was adopted by using a frozen ResNet-50 as a feature extractor and a linear Support Vector Machine (SVM) as the classifier. This design improved classification accuracy to above 70%, showing that hybrid modelling can be more suitable when the training data are limited.

In the second stage, the study was extended to the KTH-TIPS2-b dataset under a strict Leave-One-Sample-Out protocol, where the main challenge shifted from small-sample learning to cross-domain generalization. A bilinear CNN (B-CNN) achieved 86.03% accuracy but exhibited noticeable confusion between visually similar classes such as cotton and wool. A Swin-Tiny model improved the recognition of difficult categories and reached 87.46% accuracy. The best result was obtained by Swin-Base with 384×384 input resolution, together with AutoAugment, Mixup, and gradient accumulation, achieving 90.57% single-crop accuracy without relying on multi-crop evaluation. In addition, Aluminium Foil, Lettuce Leaf, and White Bread achieved F1-scores of 100%, indicating strong separability for these categories.

Overall, the results suggest that the most suitable texture recognition strategy depends on the data regime and task difficulty. Hybrid CNN–SVM models are effective in low-data settings, whereas hierarchical vision Transformers provide stronger performance under more challenging cross-domain conditions. The findings of this study offer a practical reference for the design of real-time industrial texture recognition systems and provide a foundation for future work on lightweight deployment and edge-side implementation.

Keywords: texture recognition, industrial inspection, hybrid learning, support vector machine, convolutional neural network, vision Transformer, Swin Transformer, domain shift, real-time classification

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LIST OF SYMBOLS AND ABBREVIATIONS

B-CNN	Bilinear Convolutional Neural Network
DeepTEN	Deep Texture Encoding Network
GLCM	Gray-Level Co-occurrence Matrix
LOSO	Leave-One-Sample-Out
ONNX	Open Neural Network Exchange
SVM	Support Vector Machine
ViT	Vision Transformer
Swin	Shifted Window Transformer
x	Input image
y	Ground-truth class label
X	Image/sample space
Y	Label space
f	Classification function mapping input images to labels
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DL	Deep Learning
DTD	Describable Textures Dataset
F1-score	Harmonic mean of precision and recall
FPS	Frames Per Second
GPU	Graphics Processing Unit
KTH-TIPS2-b	KTH Textures under varying Illumination, Pose and Scale 2-b
ML	Machine Learning
MobileNetV2	MobileNet Version 2
ResNet	Residual Network

CHAPTER 1 Introduction

1.1 Background

Texture recognition is a fundamental topic in computer vision and has long been studied in areas such as material analysis, surface understanding, scene interpretation, and industrial inspection. Unlike object recognition, which usually depends on overall shape and semantic structure, texture recognition focuses on repeated local patterns, surface statistics, and fine-scale appearance variations. This difference makes texture recognition particularly relevant to industrial applications, where the objective is often to determine whether a surface exhibits the expected material characteristics or quality condition rather than to identify a complete object category.

In camera-based industrial environments, especially in semiconductor manufacturing and precision surface inspection, reliable texture recognition can support automated quality control, material classification, and early defect screening. In these settings, models must often distinguish between highly similar visual patterns while remaining robust to changes in illumination, scale, viewpoint, and imaging noise. Therefore, texture recognition is not only a theoretical vision problem but also a practical engineering task with direct implications for intelligent manufacturing systems.

Traditional texture analysis methods mainly relied on handcrafted descriptors, such as statistical features, filter banks, and local pattern operators. These approaches can be effective when the imaging conditions are carefully controlled, but their flexibility is usually limited when the data distribution becomes more complex. With the development of machine learning and deep learning, texture recognition has increasingly shifted towards data-driven feature learning. Convolutional neural networks (CNNs) have become widely used because they can learn hierarchical visual representations directly from images. However, texture images often contain weak structural cues and repeated micro-patterns, which means that standard object-oriented CNN pipelines may not always be sufficient.

To improve texture modelling, researchers have proposed a range of specialized architectures, including bilinear pooling networks and other feature aggregation methods that capture higher-order statistics. More recently, vision Transformers have attracted attention because self-attention can model long-range dependencies and global context more effectively than conventional convolutions. Among them, the Swin Transformer is especially attractive for practical vision tasks because it introduces hierarchical feature extraction and window-based attention, offering a balance between performance and computational cost.

Although the available model families are increasingly diverse, selecting an appropriate architecture for real-time industrial texture recognition remains difficult. Stronger models do not always perform better in practice, especially when the dataset is small or when the testing conditions differ from the training distribution. For this reason, it is necessary to study not only final benchmark performance, but also the suitability of different modelling strategies under realistic constraints.

1.2 Motivation

This project is motivated by both practical and methodological considerations.

From a practical perspective, industrial inspection systems are expected to satisfy several requirements at the same time. They should achieve strong recognition performance, produce balanced results across categories, and remain robust under image noise and domain variation. At the same time, they should be efficient enough to support real-time or near-real-time processing in camera-based inspection pipelines. In real deployment scenarios, a model cannot be judged only by its peak accuracy. Its confusion pattern, computational cost, and stability under changing conditions are equally important.

From a methodological perspective, texture recognition provides a useful setting for examining how different learning paradigms behave under different data conditions. In particular, two issues are common in practice. The first is limited training data. Texture datasets often contain relatively few images per class, making large end-to-end networks prone to overfitting. The second is domain shift. Even when the dataset size is more reasonable, appearance statistics can vary noticeably across samples, making generalization difficult. These challenges mean that the best-performing solution may not always be a single end-to-end deep network.

A further motivation for this thesis is to document the reasoning behind model evolution. In many engineering projects, the final model is presented without fully explaining why earlier designs were insufficient. In this study, however, the progression from conventional CNNs to a hybrid CNN–SVM strategy and then to Swin-based

Transformers forms a central part of the research contribution. Each transition was driven by an observed limitation in the previous stage. This makes the work suitable not only as an experimental comparison, but also as a structured investigation into how model choice should adapt to the underlying data regime.

1.3 Problem Statement

This thesis addresses the problem of how to design and evaluate texture recognition models for real-time camera-based industrial inspection under two main constraints: limited labelled data and domain shift.

The first challenge is the small-sample problem. When only a small number of labelled images are available for each category, large neural networks can easily memorize the training set instead of learning generalizable texture representations. This typically leads to a large gap between training and validation performance. In industrial applications, collecting large-scale, well-balanced texture datasets is often costly and time-consuming, so this issue is highly relevant in practice.

The second challenge is domain shift. Even when the number of samples is larger, images from different physical instances of the same material may still vary substantially. A model trained on one subset of samples may therefore perform poorly when evaluated on previously unseen subsets. This problem becomes more evident under strict evaluation protocols designed to measure cross-sample generalization.

These challenges raise an important question: which modelling strategy is most appropriate for practical texture recognition? Should one rely on end-to-end CNNs, hybrid pipelines that combine deep feature extraction with classical machine learning, or more recent attention-based Transformer architectures? This thesis seeks to answer this question through a comparative study across multiple datasets, model families, and evaluation metrics.

1.4 Aim and Objectives of the Study

The main aim of this thesis is to develop a comparative understanding of texture recognition methods for real-time industrial inspection, with particular focus on the transition from hybrid machine-learning/deep-learning models to vision Transformer architectures.

To achieve this aim, the study has the following objectives:

1. To examine the performance of conventional CNN-based models under small-sample texture classification settings and identify the main causes of overfitting.
2. To investigate whether a hybrid strategy based on deep feature extraction and SVM classification can provide a more stable solution when the training data are limited.
3. To compare texture-specialized deep architectures and hierarchical Transformer models under stricter cross-domain evaluation conditions.
4. To assess model behaviour not only by accuracy, but also by precision, recall, F1-score, and confusion matrices.
5. To discuss the trade-off between recognition performance, robustness to noise, and real-time feasibility.
6. To derive a practically meaningful recommendation within the broader context of the joint final year project.

1.5 Scope of the Joint Project and Individual Contribution

This thesis is part of a joint final year project on real-time texture recognition for industrial inspection. While the broader project considers multiple modelling directions, the present thesis focuses specifically on the evolution from conventional CNNs to hybrid learning strategies and vision Transformers. The purpose is to understand how these approaches perform under different data regimes and how their strengths and limitations relate to practical deployment needs.

The first part of the study uses the Describable Textures Dataset (DTD) to investigate the small-sample setting. A series of experiments was conducted to examine overfitting and possible remedies. End-to-end training with ResNet-50 resulted in severe overfitting, with training accuracy reaching 82% while validation accuracy remained around 65%. Subsequent attempts using lighter backbones and stronger regularization reduced overfitting to some extent, but did not provide satisfactory generalization. MobileNetV2 combined with Mixup removed the strong overfitting behaviour, but the model converged to a lower level of performance, suggesting underfitting. Based on these observations, a hybrid approach was introduced by freezing a ResNet-50 backbone as a feature extractor and training a linear SVM on the extracted features. This method increased the classification accuracy to above 70%, showing that hybrid learning can be effective when end-to-end optimization is unstable.

The second part of the study uses the KTH-TIPS2-b dataset to investigate cross-domain generalization. This dataset provides a more material-focused benchmark and is evaluated under a strict Leave-One-Sample-Out protocol. In this stage, the study compares a bilinear CNN and Swin Transformer variants. The bilinear CNN achieved 86.03% accuracy but showed noticeable confusion between cotton and wool, which are visually similar classes. Swin-Tiny improved the performance to 87.46%. The best result was achieved by Swin-Base with 384×384 input resolution, AutoAugment, Mixup, and gradient accumulation, reaching 90.57% accuracy without using multi-crop testing. This result represents the strongest model performance obtained in the present study.

Within the joint project, the collaborating work places greater emphasis on other lightweight solutions. In contrast, this thesis mainly contributes the comparative analysis of hybrid learning and Transformer-based recognition, together with a detailed discussion of how model choice should be adjusted according to data limitations and generalization requirements.

1.6 Significance of the Study

The significance of this study lies in both its practical value and its methodological contribution.

From a practical perspective, the study addresses a real need in industrial inspection, namely the ability to classify subtle surface textures reliably under limited data and changing conditions. By comparing different model families using multiple evaluation metrics, the thesis provides a more complete understanding of model behaviour than accuracy alone. This is useful for selecting models in real applications, where confusion between specific classes may matter as much as the overall score.

From a methodological perspective, the study shows that model effectiveness is closely related to the data regime. In low-data settings, hybrid CNN–SVM pipelines can provide a more reliable alternative to end-to-end deep fine-tuning. In more demanding cross-domain settings, hierarchical Transformers show stronger representation ability and better generalization. This conclusion is important because it suggests that model selection should be based on the nature of the problem rather than on architectural popularity alone.

In addition, the study demonstrates the value of iterative model analysis. The final results were not obtained by selecting a single architecture from the beginning, but by examining the limitations of each stage and refining the modelling strategy accordingly. This process reflects the reality of applied computer vision research and strengthens the practical relevance of the thesis.

1.7 Organization of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 introduces the theoretical background for the study, including machine learning and deep learning classifiers, deep feature extraction, and the attention mechanisms used in vision Transformers, with emphasis on the Swin architecture.

Chapter 3 reviews previous work related to texture recognition, including classical texture methods, texture-oriented deep architectures such as bilinear CNNs and encoding-based models, and hybrid pipelines that combine deep features with conventional classifiers.

Chapter 4 presents the project design, including the overall experimental framework, the regularization strategies used to address overfitting, the CNN–SVM hybrid design, the Swin-Base 384 configuration, and the joint comparative evaluation framework.

Chapter 5 reports the experimental results. It discusses the success of the hybrid strategy on DTD, the performance of Swin-based models on KTH-TIPS2-b, the confusion-matrix analysis of difficult categories, and the trade-off among accuracy, robustness, and real-time feasibility.

Chapter 6 outlines possible future work, especially in relation to model compression, lightweight redesign, and deployment on edge hardware.

Chapter 7 concludes the thesis by summarizing the main findings and presenting the final recommendation of the study.

CHAPTER 2 Background

2.1 Introduction

Texture recognition has developed from traditional handcrafted-feature pipelines to deep learning and, more recently, to attention-based vision architectures. Since this thesis studies the progression from hybrid machine-learning/deep-learning models to vision Transformers, it is necessary to first establish the technical background behind these approaches. This chapter therefore introduces the main concepts that support the later experimental design and analysis.

The discussion begins with the role of machine learning and deep learning classifiers in image recognition, with particular attention to their differences in feature representation and decision making. It then explains how deep convolutional models extract texture-related features and why such features can be reused in hybrid classification pipelines. Finally, the chapter introduces the attention mechanism in vision Transformers and focuses on the Swin Transformer, which serves as the strongest model family evaluated in this study. Through these sections, the chapter provides the conceptual foundation for understanding why different model types behave differently under small-sample and cross-domain texture recognition settings.

2.2 Texture Recognition as a Visual Classification Problem

Texture recognition is a visual classification task in which the goal is to assign an input image to a texture or material category according to its surface appearance. In contrast to general object recognition, texture recognition often relies less on global semantic structure and more on local repetition, micro-patterns, frequency characteristics, and statistical regularities across the image. This view is closely related to classical statistical and local-pattern texture representations such as GLCM and LBP [1], [2]. For this reason, a successful texture classifier must be able to distinguish subtle differences in spatial arrangement, edge density, granularity, reflectance pattern, and fine-scale appearance.

In practical applications, texture recognition may be applied to material identification, defect screening, fabric analysis, food inspection, and surface-quality monitoring. In industrial scenarios, the visual differences between categories are often small, while intra-class variation may still be substantial due to lighting, scale, viewpoint, and physical differences between samples. This makes texture recognition a challenging task even for modern deep models.

From a machine learning perspective, texture recognition can be described as a mapping from an image space to a finite label space. Let $x \in \mathcal{X}$ denote an input image and $y \in \mathcal{Y}$ denote its class label. The objective is to learn a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that generalizes well to unseen examples. The quality of this function depends not only on the classifier itself, but also on how the visual representation is constructed. Historically, this representation was manually designed; in modern approaches, it is usually learned directly from data.

Because texture recognition depends strongly on representation quality, the task provides a suitable context for comparing handcrafted, hybrid, convolutional, and attention-based methods. The central question is not simply which classifier is used, but rather how the image features are encoded and whether those features remain stable under data scarcity and domain variation.

2.3 Machine Learning and Deep Learning Classifiers

2.3.1 Classical Machine Learning Classifiers

Before the widespread adoption of deep learning, most texture recognition systems followed a two-stage pipeline. In the first stage, visual features were extracted from the image using handcrafted descriptors. In the second stage, these features were fed into a classifier such as Support Vector Machine (SVM), k-Nearest Neighbours (k-NN), decision trees, or random forests. In such pipelines, feature representation and classification were treated as separate components.

Among these methods, the Support Vector Machine has been especially influential in image classification [3]. SVM seeks a decision boundary that maximizes the margin between classes in a feature space. For linearly separable data, this corresponds to learning a hyperplane that best separates the samples. For more complex cases, kernel functions can be introduced to map the data into a higher-dimensional space. In practice, linear SVMs remain widely used when the extracted features are already highly informative, because they are computationally efficient and often generalize well under limited data.

The main advantage of traditional machine learning classifiers is that they can perform robustly even when the number of training samples is relatively small, provided that the feature representation is discriminative. Their weakness, however, lies in their dependence on manually designed or externally provided features. If the feature extractor fails to capture the necessary texture information, the classifier cannot compensate for that deficiency.

2.3.2 Deep Learning Classifiers

Deep learning changes this paradigm by integrating feature learning and classification into a single trainable model. Instead of relying on handcrafted descriptors, deep neural networks learn hierarchical representations directly from raw images through end-to-end optimization. This allows the system to discover task-relevant patterns automatically, often producing stronger performance than traditional pipelines when sufficient data are available.

In image recognition, deep learning classifiers are usually built upon convolutional neural networks (CNNs) or, more recently, Transformer-based architectures. These models consist of multiple layers that gradually transform the input image into a higher-level feature representation, followed by a classification head that outputs class probabilities. The learning process is typically driven by minimizing a classification loss, such as cross-entropy, through gradient-based optimization.

The strength of deep learning lies in its representation capacity. Early layers tend to capture low-level structures such as edges, corners, and simple textures, while deeper layers encode more abstract and task-specific visual patterns. However, this capacity comes at a cost. Deep models often require large amounts of data to generalize reliably, and their training may be unstable when the dataset is small or when the class distribution is difficult. In texture recognition, where the available data may be limited and inter-class differences subtle, this issue becomes particularly important.

2.3.3 Comparison Between the Two Paradigms

The main distinction between classical machine learning and deep learning lies in the relationship between representation learning and classification. In classical pipelines, feature extraction is fixed or manually engineered, and the classifier operates on those features afterward. In deep learning, feature extraction and classification are optimized jointly.

This difference leads to different strengths. Classical classifiers such as SVM are often more stable in low-data settings and can work effectively with strong pre-extracted features. Deep models, on the other hand, are more flexible and can learn more powerful representations, but they are also more sensitive to training conditions, model size, and data quality. Consequently, the choice between these paradigms should depend on the underlying task and dataset regime rather than on a simple assumption that end-to-end deep learning is always superior.

2.4 Deep Feature Extraction for Texture Analysis

2.4.1 Hierarchical Feature Learning in CNNs

Convolutional neural networks are based on local receptive fields, weight sharing, and hierarchical processing. A convolution layer applies a bank of learnable filters to the input image or feature map, producing response maps that emphasize particular local structures. As the network deepens, these responses are combined to form increasingly abstract representations.

This hierarchical property is especially relevant to texture analysis. Texture images contain repeated local cues that can often be captured well by convolutional operations. Early CNN layers respond to primitive structures such as edges, lines, and frequency-like patterns, while middle layers combine these cues into more complex motifs. In many cases, such features already contain useful information for distinguishing materials or surface categories.

Pooling and downsampling further help the model gain robustness to small spatial variations. At the same time, non-linear activation functions allow CNNs to approximate complex mappings between image patterns and labels. Well-known architectures such as VGG, ResNet, and MobileNet differ in their internal design, but they all follow this broad principle of progressively extracting informative visual features through multiple processing stages.

2.4.2 Residual Learning and Representative CNN Backbones

Among CNN architectures, residual networks are particularly important because they address the degradation problem that arises when networks become deeper [6]. A residual block introduces an identity shortcut connection, allowing the model to learn a residual function rather than a full transformation from scratch. This makes optimization easier and enables the construction of deeper networks such as ResNet-18 and ResNet-50.

ResNet-50 is widely used as a feature extraction backbone because it offers strong representational power while remaining manageable in practice. Its depth allows it to capture rich image patterns, including fine-grained local features relevant to texture recognition. However, when the training set is small, such a model may overfit during end-to-end fine-tuning.

MobileNetV2 represents a different design philosophy. It is intended to be lightweight and efficient, using depthwise separable convolutions, inverted residual blocks, and linear bottlenecks to reduce computation [7]. Although such a design is attractive for real-time systems, its lower capacity may also limit performance if the task demands highly discriminative representations. Therefore, different CNN backbones offer different trade-offs between capacity, efficiency, and generalization.

2.4.3 Feature Transfer and Reuse

A major advantage of CNNs is that the representations learned by large-scale pretraining can often be transferred to new tasks. Large-scale datasets such as ImageNet enabled the development of generic visual representations [4], and early deep feature studies such as DeCAF showed that pretrained CNN activations could be reused effectively for downstream recognition tasks [5]. In transfer learning, a backbone pretrained on a large dataset is adapted to a smaller target dataset, either by fine-tuning all layers or by using the network as a fixed feature extractor. This is particularly useful in texture recognition, where dataset sizes are often modest.

When a pretrained CNN is used as a feature extractor, the output of an intermediate or final feature layer can be treated as a compact numerical representation of the image. These deep features can then be passed to another classifier, such as an SVM. In this sense, the CNN no longer acts as the entire recognition system, but as a learned feature encoder.

This approach is important for the present thesis because it forms the basis of hybrid learning. Instead of relying on end-to-end fine-tuning in a challenging small-data regime, it is possible to use the CNN to extract stable visual features and then apply a simpler classifier to make the final decision. This can reduce optimization difficulty while still benefiting from the expressive power of deep representations.

2.5 Hybrid Deep Learning and Machine Learning Strategies

A hybrid model combines the representation strength of deep learning with the decision stability of classical machine learning. In image classification, the most common form of such a strategy is to use a pretrained or partially frozen CNN as a feature extractor and then train a machine learning classifier, often an SVM, on top of the extracted features [3], [5].

The rationale for this design is straightforward. Deep networks are powerful at learning general-purpose visual representations, especially when pretrained on large image datasets. However, when the target dataset is small, end-to-end optimization may become unstable and lead to overfitting. A classical classifier, by contrast, can often operate more reliably once a strong feature space has been obtained. Therefore, the hybrid pipeline separates the representation problem from the final decision boundary problem.

This separation offers several benefits. First, it reduces the number of trainable parameters in the downstream task, which can improve stability under limited supervision. Second, it allows the use of classifiers with strong margin-based generalization properties. Third, it makes the training pipeline easier to interpret, because the feature extractor and classifier can be studied independently.

However, hybrid models also have limitations. Since the classifier is trained on fixed features, it cannot jointly refine the feature representation according to the final objective. As a result, hybrid models may be less flexible than end-to-end models when abundant data are available. Their usefulness is therefore most pronounced in settings where data are scarce, optimization is difficult, or model simplicity is desirable.

In the context of this thesis, the hybrid paradigm is important not only as a practical solution but also as a conceptual bridge between classical machine learning and end-to-end deep learning. It shows that these two traditions are not strictly opposed. Instead, they can complement each other when the dataset conditions make fully end-to-end learning suboptimal.

2.6 Vision Transformers for Image Recognition

2.6.1 From Sequence Modelling to Vision

Transformers were originally developed for sequence modelling in natural language processing. Their core component is the self-attention mechanism, which allows each element in a sequence to interact with all other elements and weigh their importance adaptively. This design enables the model to capture long-range dependencies more effectively than purely recurrent or locally constrained architectures.

The success of Transformers in language processing motivated researchers to adapt the same idea to image recognition. In a Vision Transformer, an image is divided into a sequence of patches, and each patch is embedded into a vector representation [14]. These patch embeddings are then processed by Transformer blocks, allowing the model to learn relationships across the entire image through self-attention.

This design differs fundamentally from CNNs. Convolution is inherently local, with spatial context expanding gradually through depth. Self-attention, in contrast, can connect distant regions directly in a single layer. For texture recognition, this property is attractive because texture cues may be distributed across the image, and long-range consistency can matter in addition to local detail.

2.6.2 Self-Attention Mechanism

The self-attention mechanism operates by projecting each input token into three vectors: query, key, and value. The similarity between the query of one token and the keys of other tokens determines the attention weights, and these weights are used to aggregate the corresponding value vectors. In simplified form, this process can be written as

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V$$

where Q, K, and V denote the query, key, and value matrices, and d is a scaling factor related to feature dimension.

Through this mechanism, the model can determine which parts of the image are relevant to each token representation. Multi-head attention extends this idea by allowing several attention operations to be learned in parallel, so that different relational patterns can be captured simultaneously.

For vision tasks, self-attention provides a flexible way to model global context. However, the standard formulation also introduces high computational cost, because attention is computed between all pairs of tokens. This makes naive global self-attention expensive, especially for high-resolution images.

2.6.3 Vision Transformer Characteristics

Vision Transformers have shown strong performance in many image classification tasks because they can learn powerful global representations [14]. However, compared with CNNs, they often depend more heavily on pretraining, data scale, and training strategy. Without sufficient data or suitable regularization, they may not generalize as effectively as convolutional architectures.

Another notable difference is the inductive bias. CNNs naturally encode locality and translation-related structure through convolution and pooling. Vision Transformers are more flexible but less constrained, which can be advantageous when large-scale data are available, yet challenging when training data are limited. For this reason, later Transformer variants introduced hierarchical design and locality-aware mechanisms to improve practical performance.

2.7 Swin Transformer

2.7.1 Motivation for Hierarchical Vision Transformers

The standard Vision Transformer applies global attention across all image patches, which can be computationally demanding and may not align well with the multi-scale structure of natural images. Many visual tasks benefit from hierarchical feature representations, where low-level and high-level information are processed at different resolutions. This is a natural property of CNNs, but not of the original flat Transformer design.

The Swin Transformer was proposed to address this issue by introducing a hierarchical architecture together with shifted-window self-attention [15]. Instead of computing attention over the entire image, Swin restricts self-attention to non-overlapping local windows. This greatly reduces computational cost. To allow information exchange across windows, the window partition is shifted between consecutive blocks. As a result, the model can gradually build broader contextual understanding while maintaining efficiency.

2.7.2 Architecture of Swin Transformer

The Swin Transformer begins by partitioning the image into non-overlapping patches and projecting them into an embedding space. These patch embeddings then pass through several stages. Within each stage, Transformer blocks operate on local windows, and patch merging layers reduce spatial resolution while increasing channel dimension. In this way, the network forms a hierarchical feature pyramid, similar in spirit to CNNs.

A key design component is the alternation between regular window-based multi-head self-attention and shifted-window multi-head self-attention. Regular windows keep the computation local and efficient. Shifted windows enable neighbouring windows from the previous layer to interact, which improves cross-window communication. This design preserves much of the efficiency advantage of local attention while still allowing broader feature integration over depth.

For image classification, the final-stage features are aggregated and passed to a linear head to produce class predictions. Different model sizes, such as Swin-Tiny and Swin-Base, vary in depth, width, and computational complexity. Larger variants usually offer stronger representational power, while smaller variants are more lightweight and efficient.

2.7.3 Relevance of Swin Transformer to Texture Recognition

The Swin Transformer is particularly relevant to texture recognition for several reasons. First, its hierarchical structure allows it to capture both fine local details and broader spatial relationships [15]. This is important because textures are defined not only by local micro-patterns but also by their arrangement and consistency across the image. Second, the shifted-window mechanism improves contextual aggregation without incurring the full cost of global attention, which is beneficial for real-time or high-resolution applications. Third, the architecture is flexible enough to benefit from modern training strategies such as data augmentation, label smoothing, and Mixup [8].

In material and texture classification, subtle differences between visually similar classes may depend on both local structure and global context. A purely local model may overlook image-wide consistency, while a purely global model may lose fine detail or become too expensive. Swin provides a balanced solution by combining locality, hierarchy, and attention-based modelling. For this reason, it serves as the main high-performance architecture examined in the later experimental chapters of this thesis.

2.8 Evaluation Metrics Relevant to Texture Recognition

Although the detailed experimental setup is presented in later chapters, several evaluation metrics should be introduced here because they are fundamental to understanding model behaviour.

Accuracy measures the proportion of correctly classified samples among all samples. It provides a simple overall indicator of performance, but it does not reveal whether certain classes are much easier or harder than others.

Precision measures the proportion of correctly predicted positive samples among all samples predicted as a given class. High precision indicates that the model makes relatively few false positive predictions for that category.

Recall measures the proportion of correctly predicted positive samples among all actual samples of a given class. High recall means that the model successfully retrieves most samples belonging to that category.

F1-score is the harmonic mean of precision and recall. It is useful when a balanced measure is required, especially in multi-class problems where class-level performance may vary.

Confusion matrix analysis shows how predictions are distributed across true and predicted classes. For texture recognition, this is particularly informative because many errors arise from confusion between visually similar categories rather than random failure.

In this thesis, these metrics are used together because no single measure is sufficient to describe model quality in practical inspection scenarios. A model with high overall accuracy may still be unsuitable if it consistently confuses critical material categories.

2.9 Real-Time Considerations and Robustness to Noise

Because this project is motivated by real-time industrial inspection, model evaluation cannot be based solely on recognition accuracy. The computational cost of a model also matters. In practice, inference speed depends on factors such as input resolution, parameter count, architectural design, memory access pattern, and deployment hardware. Lightweight CNNs are often attractive because they reduce computation, but they may lose discriminative power. Larger Transformers may achieve stronger performance, but their latency and memory requirements can be more demanding.

Noise robustness is another important consideration. Industrial images may be affected by sensor noise, illumination fluctuations, blur, compression artefacts, or surface contamination. A practical texture recognition model should therefore maintain stable predictions even when image quality is imperfect. The relationship between robustness and model design is complex. Some models benefit from stronger contextual modelling, while others rely more on local signal quality. Training strategies such as augmentation and regularization can also influence robustness.

For these reasons, the later chapters of this thesis examine not only the absolute recognition performance of different architectures, but also the trade-off between accuracy, real-time feasibility, and robustness. This broader evaluation perspective is necessary for meaningful deployment-oriented model selection.

2.10 Chapter Summary

This chapter has introduced the main technical background required for the rest of the thesis. It first explained texture recognition as a visual classification problem and outlined the distinction between classical machine learning classifiers and deep learning models. It then discussed deep feature extraction in CNNs and showed how such features can be reused in hybrid learning pipelines. After that, the chapter introduced vision Transformers and the self-attention mechanism, followed by a focused explanation of the Swin Transformer and its relevance to texture recognition. Finally, it reviewed the evaluation metrics and practical considerations that are important for real-time industrial inspection.

These concepts form the basis for the subsequent chapters. Chapter 3 will build upon this background by reviewing existing studies on texture recognition, including classical methods, texture-specialized deep models, and hybrid or Transformer-based approaches.

CHAPTER 3 Literature Review

3.1 Introduction

Texture recognition has been studied for several decades and has evolved through several distinct methodological stages. Early research was dominated by handcrafted descriptors designed to capture statistical or structural properties of texture images. Later, machine learning classifiers were introduced to improve decision boundaries on top of these descriptors. With the rise of deep learning, texture recognition shifted toward convolutional feature learning, orderless pooling, and texture-specialized deep architectures. More recently, vision Transformers have emerged as an alternative representation paradigm, offering stronger global modelling ability and hierarchical context aggregation.

The purpose of this chapter is to review this evolution and identify the main research trends relevant to the present thesis. Rather than attempting an exhaustive survey of all texture-recognition studies, this chapter focuses on the lines of work that are most closely connected to this project: classical texture descriptors, CNN-based representations, texture-specialized deep models, hybrid deep-feature-plus-classifier pipelines, and Transformer-based visual architectures. The chapter concludes by clarifying the research gap addressed in this thesis and positioning the present study within the existing literature.

3.2 Classical Texture Representation Methods

Early texture recognition methods were largely based on handcrafted representations. One of the most influential early directions was statistical texture analysis. Haralick et al. introduced gray-level co-occurrence matrix (GLCM) features [1], which describe spatial dependencies between pixel intensities and derive statistics such as contrast, correlation, homogeneity, and angular second moment. These features established an important foundation for later texture analysis because they formalized the idea that texture can be characterized through local intensity relationships rather than object shape alone.

Another influential family of methods relied on local pattern encoding. Among these, Local Binary Patterns (LBP) became one of the most widely used handcrafted descriptors for texture classification. Ojala et al. proposed a multiresolution and rotation-invariant LBP framework that encodes the local neighbourhood of a pixel into a binary pattern and summarizes the image through histograms of these patterns [2]. The popularity of LBP was largely due to its simplicity, computational efficiency, and strong empirical performance on texture benchmarks. For many years, it served as a standard baseline in both academic and applied texture-recognition tasks.

Although handcrafted descriptors achieved meaningful progress, they also had clear limitations. Their performance often depended heavily on careful parameter design and they could struggle when appearance variation became more complex, especially under realistic changes in illumination, scale, viewpoint, and clutter. As material recognition tasks moved from controlled laboratory datasets toward more challenging real-world scenarios, the need for stronger and more adaptive representations became increasingly evident. This limitation motivated the transition from fixed descriptors to learned visual features.

3.3 Benchmark Datasets and the Shift Toward More Realistic Evaluation

The development of texture-recognition methods has been closely tied to the datasets used for evaluation. Earlier benchmarks often focused on relatively controlled acquisition settings, which made it easier for handcrafted methods to perform well. However, as the field progressed, newer datasets were introduced to better reflect real visual variability. KTH-style material categorisation studies helped establish material recognition as a challenging problem involving variations in illumination, pose, and scale [10]. Such datasets became important for studying generalization across physical samples rather than only memorization of repeated patterns.

A particularly important step in this evolution was the introduction of the Describable Textures Dataset (DTD). Cimpoi et al. proposed DTD to support recognition of describable texture attributes collected in the wild, rather than only recognizing controlled material exemplars [9]. DTD helped shift the field toward more semantic and visually diverse texture understanding. These benchmarks collectively encouraged the transition from handcrafted texture analysis to data-driven representation learning.

The literature therefore suggests that benchmark difficulty is not only a matter of class number, but also of evaluation protocol and domain variation. Controlled datasets are useful for testing feature sensitivity, whereas stricter protocols and more realistic datasets are better for measuring generalization. This distinction is directly relevant to the present thesis, which compares models under both a small-sample semantic texture setting and a stricter cross-sample material-recognition setting.

3.4 CNN-Based Texture Representation

The rise of deep learning changed the texture-recognition literature substantially. Instead of relying on manually designed features, convolutional neural networks made it possible to learn representations directly from image data. A key early observation was that CNN features pretrained on large-scale object-recognition datasets such as ImageNet could serve as strong generic image descriptors across multiple recognition tasks [4]. Donahue et al. showed through DeCAF that deep convolutional activation features were effective for generic visual recognition, suggesting that deep representations could generalize beyond the specific object categories used during pretraining [5]. This result encouraged the use of CNN backbones as general-purpose feature extractors in texture and material recognition.

However, texture recognition differs from object recognition in an important way: texture often depends less on global shape and more on distributed local patterns. This led researchers to question whether standard CNN pipelines designed for object categories were sufficiently well aligned with texture-specific properties. In response, studies began exploring the use of convolutional layers as “deep filter banks” rather than relying only on fully connected outputs. This shift emphasized the value of local convolutional responses and orderless pooling for texture description.

Cimpoi et al. proposed Deep Filter Banks and the FV-CNN representation, which treated convolutional activations as local descriptors and pooled them using Fisher Vectors [11]. Their results showed that orderless pooling of convolutional features could outperform more conventional CNN classification pipelines on several texture, material, and scene-recognition tasks. This was a major result for the field because it demonstrated that deep convolutional representations were highly useful for texture analysis, but also that texture recognition benefited from pooling strategies different from those commonly used in object classification.

The broader implication of this line of work is that CNNs did not simply replace handcrafted texture features in a one-to-one fashion. Rather, they changed the representation space while preserving an important insight from earlier texture literature:

orderless aggregation of local descriptors remains highly relevant. This idea later influenced several texture-specialized deep architectures, including bilinear pooling and encoding-based models.

3.5 Texture-Specialized Deep Architectures

As CNN-based recognition matured, researchers proposed architectures designed more specifically for fine-grained and texture-rich visual tasks. One important example is the Bilinear CNN (B-CNN) introduced by Lin et al. [12]. Bilinear CNNs combine two feature extractors through a pooled outer product, thereby capturing pairwise local feature interactions in an orderless manner. Although originally introduced for fine-grained recognition, the model was also shown to generalize several orderless pooling methods used in texture and region description. This made B-CNN an especially relevant architecture for texture recognition, where subtle local interactions often matter more than global shape.

Another influential model is DeepTEN, proposed by Zhang et al. [13]. DeepTEN introduced an Encoding Layer that integrates dictionary learning and residual encoding into an end-to-end trainable network. In contrast to modular pipelines that separate feature extraction from pooling, DeepTEN learns both jointly, allowing the representation to be optimized specifically for material and texture recognition. The authors reported strong results on texture and material benchmarks including KTH-TIPS-2b and MINC, showing that encoding-based deep aggregation could outperform more conventional CNN baselines.

These models represent an important stage in the literature because they show that texture recognition benefits from more than simply making CNNs deeper. What matters is the way local features are aggregated. Standard classification heads may overlook the statistical and orderless nature of texture, whereas bilinear pooling and encoding layers preserve richer information about repeated local structure. For this reason, texture-specialized deep models occupy an important middle ground between generic CNN pipelines and more recent Transformer-based solutions.

At the same time, these architectures also have practical limitations. Bilinear and higher-order pooling methods can be computationally expensive or feature-dimension intensive, while end-to-end encoding-based models still depend on sufficient training data and careful optimization. As a result, the literature does not support the view that texture-specific deep architectures are universally superior under all conditions. Instead, their effectiveness depends on the interaction between data regime, task difficulty, and computational budget.

3.6 Hybrid Deep Learning and Classical Machine Learning

Alongside end-to-end deep models, another line of literature explored hybrid strategies that combine deep features with classical machine learning classifiers. The broader theoretical basis for this approach was reinforced by work showing that pretrained CNN features can function as strong generic image descriptors [5]. Once such a feature space is available, a simpler classifier such as an SVM may be sufficient to construct an effective decision boundary, especially when labelled training data are limited [3].

This idea is particularly relevant to texture recognition. Texture datasets are often smaller than large-scale object benchmarks, and the risk of overfitting is therefore more severe when large CNN backbones are fine-tuned end to end. Hybrid pipelines address this issue by separating representation learning from classification: the CNN is used as a frozen or partially frozen feature extractor, while the final classifier is trained independently. In low-data settings, this can reduce optimization instability and limit the number of task-specific parameters that must be learned.

The literature on hybrid texture recognition is less dominant than the literature on end-to-end deep learning, but it remains conceptually important. It highlights that the best recognition strategy may depend on the data regime rather than on architectural novelty alone. In particular, when the available training data are insufficient to support reliable fine-tuning of a high-capacity backbone, classical margin-based classifiers can still offer competitive performance once strong deep features have been extracted. This view is closely aligned with the present thesis, which treats the hybrid CNN–SVM pipeline not as a temporary workaround, but as a valid modelling choice under constrained supervision.

3.7 Vision Transformers and the Emergence of a New Representation Paradigm

A major recent development in computer vision has been the introduction of Transformer-based architectures. Dosovitskiy et al. demonstrated that images can be represented as sequences of patches and processed using the Transformer framework originally developed for natural language processing [14]. Their Vision Transformer (ViT) showed that self-attention-based models could achieve strong image-classification performance when trained at scale. This work established Transformers as a serious alternative to CNNs in visual recognition.

For texture recognition, Transformers are attractive because they model relationships between image regions more flexibly than convolutions. While CNNs build larger receptive fields progressively, self-attention can connect distant regions directly. This may be beneficial when texture categories depend not only on local micro-patterns but also on broader spatial consistency across the image. However, early Vision Transformers also had limitations, including substantial data demands and high computational cost, which made them less straightforward to apply in practical scenarios with moderate dataset sizes.

To address some of these issues, Liu et al. proposed the Swin Transformer, a hierarchical vision Transformer based on shifted-window attention [15]. By restricting attention to local windows while shifting the window partition across layers, Swin reduces computational complexity and introduces a multi-scale representation structure closer to CNN design. The model achieved strong image-classification and dense-prediction results, which made it especially attractive as a practical visual backbone. Its hierarchical structure and balance between local and global modelling are also conceptually well suited to texture recognition, where both fine detail and contextual regularity matter.

Although the Transformer literature did not originate specifically from texture classification, it has increasingly influenced the field by offering a new way to represent visual patterns. Compared with earlier CNN-based texture-specialized models, Swin suggests that part of the texture-recognition problem may be addressed through stronger feature hierarchy and context modelling rather than only through orderless pooling. This does not invalidate earlier CNN-based insights; rather, it extends them and raises the question of whether Transformer backbones can better handle visually ambiguous classes and stronger domain variation.

3.8 Summary of the Literature and Research Gap

The literature reviewed above suggests several broad conclusions. First, classical handcrafted descriptors laid the conceptual foundation of texture analysis by emphasizing local statistics and repeated pattern structure. Second, CNN-based approaches substantially improved representation quality, especially when pretrained deep features and orderless pooling were combined. Third, texture-specialized deep models such as B-CNN and DeepTEN showed that careful feature aggregation is central to high-performance texture recognition. Fourth, hybrid CNN-plus-classifier strategies remain relevant in low-data regimes, even though they are less emphasized in recent mainstream literature. Finally, Transformer-based architectures introduce a new representation paradigm with stronger global modelling and hierarchical context aggregation.

However, an important gap remains in how these approaches are compared under different data conditions. Much of the literature evaluates methods either on a single benchmark or within a single modelling paradigm. Fewer studies explicitly analyze how model preference should change between small-sample texture classification and stricter cross-domain material recognition. Likewise, not enough attention has been paid to the practical question of when a hybrid method may be more appropriate than end-to-end deep fine-tuning, or when a hierarchical Transformer may justify its additional complexity through improved generalization.

This thesis addresses that gap by providing a comparative study across two different regimes. The first is a small-sample semantic texture setting, where the main challenge is overfitting. The second is a stricter cross-sample material-recognition setting, where the main challenge is domain shift and fine-grained confusion between visually similar categories. By examining conventional CNNs, a hybrid CNN–SVM strategy, texture-specialized deep architectures, and Swin-based Transformers within a single study, this thesis aims to provide a more practically grounded understanding of model selection for real-time industrial texture recognition.

3.9 Chapter Summary

This chapter has reviewed the main literature strands relevant to the present work. It began with classical handcrafted texture descriptors, then discussed the transition toward CNN-based feature learning, orderless deep pooling, and texture-specialized deep architectures. It also examined hybrid learning strategies that combine deep feature extraction with classical classification, and finally reviewed the emergence of vision Transformers and the Swin Transformer as a promising modern backbone. Based on this review, the next chapter will describe how these ideas were translated into the project design and experimental framework of this thesis.

CHAPTER 4 Project Design

4.1 Introduction

This chapter presents the design of the proposed texture recognition study and explains how the experimental pipeline was implemented. The project was not organized as a single one-shot comparison, but as a staged investigation in which model design evolved according to the main difficulty observed at each stage. The first stage focused on the small-sample setting represented by the Describable Textures Dataset (DTD), where severe overfitting became the dominant issue. The second stage shifted to the KTH-TIPS2-b dataset, where the main challenge was no longer only data scarcity, but also cross-sample domain shift under a stricter evaluation protocol. For this reason, the overall design of the project followed a progressive logic: first to diagnose and mitigate overfitting, then to compare increasingly expressive architectures under stronger generalization requirements.

Within the broader joint final year project, the present thesis concentrates on model evolution from conventional convolutional baselines to hybrid CNN–SVM strategies and then to Transformer-based architectures. The project design therefore had two goals. The first was to identify which modelling strategy was most suitable in each data regime. The second was to establish a consistent experimental framework so that the results could later be compared in terms of accuracy, precision, recall, F1-score, confusion patterns, and practical deployment relevance.

To support this objective, all models were implemented in PyTorch or timm-based pipelines and were trained using ImageNet-pretrained backbones whenever available. The codebase included conventional CNN baselines, a hybrid feature-extraction-plus-SVM pipeline, a texture-specialized bilinear CNN, and Transformer-based models including DeiT-Small and Swin Transformer. Their designs are described in the following sections.

4.2 Overall Experimental Framework

The project design can be understood as a two-stage comparative framework.

In the first stage, DTD was used as a diagnostic benchmark for low-data learning. Since the dataset contains many categories but only a limited number of training samples per class, it was suitable for testing whether conventional end-to-end fine-tuning could generalize reliably. The initial experiments therefore used CNN-based pipelines with increasingly stronger regularization. When these strategies either failed to remove overfitting or over-corrected into underfitting, a hybrid architecture was introduced in which deep features were extracted by a pretrained CNN and classified by a linear SVM. This stage was intended to answer a practical question: when the data are insufficient for stable fine-tuning, can representation learning and classification be separated more effectively?

In the second stage, the project moved to KTH-TIPS2-b, which places greater emphasis on physical materials and cross-sample variation. Under this setting, the goal was not merely to prevent overfitting, but to evaluate whether different architectures could generalize across unseen material instances. To address this, the design compared a standard CNN baseline, a lighter CNN with stronger augmentation, a hybrid ResNet50+SVM pipeline, a texture-specialized B-CNN, a ViT-lite baseline based on DeiT-Small, and Swin Transformer variants at both 224 and 384 input resolutions. This made it possible to compare conventional local feature learning, higher-order texture aggregation, and hierarchical self-attention within a single experimental framework.

Although the thesis focuses mainly on recognition performance and model behaviour, the design was also constructed with later deployment-oriented discussion in mind. Consequently, model size, training stability, input resolution, and augmentation intensity were treated as design variables rather than fixed settings. This later supports the discussion of real-time feasibility and robustness in Chapter 5.

4.3 Dataset Preparation and Data Splits

4.3.1 DTD Setup

DTD was used as the main benchmark for the first stage of the study. The dataset contains 47 texture categories and was employed to examine the effect of limited per-class supervision on different model families [9]. In the codebase, the DTD data were organized into standard train, val, and test folders for the main CNN, MobileNetV2, ViT-lite, and hybrid CNN–SVM experiments. A preprocessing utility was also prepared to generate official fold-based splits from the original DTD labels files, ensuring reproducible dataset construction before model training.

For the main DTD experiments reported in this thesis, the data were treated according to the train/validation/test protocol used in the project, consistent with the stated 80/10/10-style experimental logic. Training data were used for model fitting, the validation split was used for checkpoint selection and hyperparameter observation, and the test split was reserved for final reporting. This separation was especially important in the small-sample setting, where performance fluctuations could otherwise be misleading.

In all DTD pipelines, images were normalized using the standard ImageNet mean and standard deviation. Validation and test transforms used deterministic resizing and center cropping when the backbone expected 224×224 input, while higher-resolution models used direct resizing to 384×384. For some auxiliary high-resolution DTD experiments, a wrapper-based random split strategy was used on the raw image folder for rapid ablation, but the core project narrative and reported conclusions remained anchored in the train/validation/test DTD setup.

4.3.2 KTH-TIPS2-b Setup

KTH-TIPS2-b was used as the second-stage benchmark because it offers a more material-oriented recognition problem with stronger variation across physical samples [10]. The dataset contains 11 classes: aluminium foil, brown bread, corduroy, cork, cotton, cracker, lettuce leaf, linen, white bread, wood, and wool. Unlike the DTD experiments, the KTH-TIPS2-b design explicitly adopted a strict cross-sample evaluation protocol.

The dataset loader was implemented to read the original folder hierarchy by material category and sample identity. In the experiments described in this thesis, sample_a, sample_b, and sample_c were used as the training set, while sample_d was used as the validation/test target. This leave-one-sample-out style design ensured that the model was evaluated on a physically different material instance from those seen during training. As a result, performance under this protocol better reflected the model's ability to generalize across domain shift rather than simply memorize sample-specific appearance statistics.

This KTH-TIPS2-b protocol was reused across the CNN, MobileNetV2, hybrid ResNet50+SVM, B-CNN, ViT-lite, Swin-Tiny, and Swin-Base experiments so that later comparisons remained directly meaningful.

4.4 Data Preprocessing and Augmentation Strategy

A central part of the project design was the adjustment of preprocessing and augmentation intensity according to model family and dataset difficulty. Because texture recognition depends strongly on fine local patterns, the augmentation strategy had to improve generalization without destroying the discriminative surface information required for classification.

For conventional CNN baselines on DTD, the training transform included random resized cropping, horizontal and vertical flips, random rotation, and color jitter. In the ResNet-based regularized baseline, the crop size was 224 and the augmentation scale was moderately constrained. Validation and test images were resized to 256 and center-cropped to 224, ensuring consistent evaluation input. This design followed a standard transfer-learning convention while introducing enough stochasticity to reduce trivial memorization.

For the MobileNetV2 experiments, the augmentation strategy was made stronger. In addition to random resized cropping, horizontal and vertical flips, random rotation, and color jitter, Random Erasing was added to simulate partial information loss and force the model to rely on broader texture cues. Mixup was also incorporated during training following the regularization strategy proposed by Zhang et al. [8]. Given two samples and their labels, Mixup constructs a virtual training example as

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

where λ is sampled from a Beta distribution. In the MobileNetV2 DTD pipeline, $\alpha = 0.4$ was used, reflecting a relatively strong interpolation regime intended to suppress overfitting.

For the KTH-TIPS2-b experiments, preprocessing was standardized more carefully across models so that architectural comparison remained fair. The ResNet18 and MobileNetV2 baselines used 224×224 input resolution. Transformer-based models were divided into two groups: ViT-lite and Swin-Tiny used 224×224 input, while the strongest Swin-Base configuration used 384×384 input. B-CNN was also evaluated with 384×384 input in order to preserve more fine-scale texture detail. In these higher-capacity pipelines, AutoAugment with the ImageNet policy was added together with random resized cropping and random flipping. This was intended to increase appearance diversity while still preserving the broader texture structure of the scene.

Across all pipelines, image normalization used the ImageNet channel mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225], ensuring compatibility with pretrained backbones.

4.5 Baseline CNN Design for Small-Sample Learning

4.5.1 ResNet-Based Baseline

The initial CNN baseline was designed to test whether a standard pretrained convolutional backbone could be adapted successfully to DTD. A ResNet-based pipeline was used because residual networks are well-established transfer-learning baselines and offer a reasonable balance between representational strength and implementation stability [6]. In the regularized ResNet18 code path, the pretrained backbone was partially frozen: the earlier layers were fixed, while layer3, layer4, and the final classification head remained trainable. The original fully connected layer was replaced by a dropout layer followed by a new linear classifier matching the target number of classes.

The training objective was cross-entropy loss with label smoothing, and the optimizer was Adam with cosine annealing learning-rate scheduling. Dropout, weight decay, data augmentation, and selective fine-tuning were all introduced to reduce overfitting. This design reflects a typical transfer-learning strategy for small datasets: freeze the lower-level general features, fine-tune the more task-specific layers, and regularize the classifier head.

However, the broader DTD baseline study also included a stronger end-to-end ResNet-50 baseline, whose behaviour established the main motivation for the hybrid stage. According to the experimental record, this larger model reached high training accuracy but generalized poorly, confirming that representational capacity alone was not enough in the low-sample regime.

4.5.2 MobileNetV2 with Stronger Regularization

The second major baseline design replaced the heavy residual backbone with MobileNetV2 [7]. The motivation was twofold. First, a lighter backbone might be less prone to memorization than a high-capacity ResNet-50. Second, MobileNetV2 is more relevant to practical deployment because of its computational efficiency.

In the implemented pipeline, a pretrained MobileNetV2 model was used, and the first ten feature blocks were frozen while the later blocks and classifier were fine-tuned. The classifier head was replaced by a dropout layer and a new linear layer. Compared with the ResNet baseline, the MobileNetV2 design relied more heavily on aggressive regularization: stronger data augmentation, Random Erasing, Mixup with $\alpha = 0.4$, Adam optimization, and cosine annealing scheduling. The intent of this design was to determine whether overfitting on DTD could be reduced by simultaneously lowering capacity and increasing augmentation diversity.

This configuration played an important role in the design logic of the project. Even though it improved training stability, it also showed that strong regularization may remove overfitting at the cost of underfitting. This observation directly motivated the next design step, namely to separate deep representation extraction from final classification.

4.6 Hybrid CNN–SVM Design

4.6.1 Motivation and Design Principle

The hybrid design was introduced to address a specific failure mode observed in the DTD experiments. End-to-end deep optimization either overfit strongly or became too constrained after aggressive regularization. This suggested that the main issue was not the absence of useful visual features, but the instability of classifier fitting under limited supervision. Therefore, the project adopted a hybrid pipeline in which the deep network was used only as a fixed representation extractor, while the final decision boundary was learned by a classical classifier.

4.6.2 ResNet50 Feature Extraction

In both the DTD and KTH-TIPS2-b hybrid experiments, a pretrained ResNet-50 backbone was loaded and truncated before the final classification layer. The remaining convolutional body, including the global pooling output, was wrapped as a sequential feature extractor. During inference over the dataset, each image was passed through this frozen network and the resulting feature tensor was flattened into a one-dimensional vector. This produced a compact deep representation for each sample while avoiding end-to-end gradient updates on the target dataset.

The use of ResNet-50 in this role was deliberate. Although the full model overfit when fine-tuned on DTD, its pretrained convolutional representation remained strong as a general-purpose visual encoder. By fixing the feature extractor, the design retained this representational benefit while removing the instability associated with updating millions of parameters on a small dataset.

4.6.3 Standardization and Linear SVM Classification

After extraction, the features were standardized using StandardScaler, which was fitted on the training features and then applied to validation and test features. This step ensured that the SVM received features on a comparable scale, which is important for margin-based classification.

The classifier itself was implemented as a linear SVM using LinearSVC with C=0.01, max_iter=10000, and dual=True. For DTD, the training and validation features were concatenated before fitting the final SVM, so that the classifier could use more labelled examples once the representation design had been fixed. For KTH-TIPS2-b, the SVM was trained on the extracted features from sample_a, sample_b, and sample_c, and then evaluated on sample_d under the same leave-one-sample-out protocol used by the deep models.

This design is methodologically important because it reinterprets the CNN not as the whole classifier, but as a learned feature bank. The SVM then becomes a lightweight decision layer with strong margin-based behaviour, which is often advantageous when the number of labelled samples is limited.

4.7 Texture-Specialized Deep Model Design

4.7.1 Bilinear CNN

To move beyond standard CNN classification heads, the project also included a texture-specialized bilinear CNN on KTH-TIPS2-b [12]. The implemented B-CNN used a pretrained ResNet-34 as the feature extractor rather than ResNet-50. This design choice was practical: the output channel dimension of ResNet-34 is 512, which yields a bilinear feature of size 512×512, whereas a ResNet-50 backbone would produce a much larger bilinear representation and a substantially heavier classifier, increasing the risk of overfitting and excessive memory use.

$F \in \mathbb{R}^{C \times HW}$ denote the reshaped convolutional feature map. Bilinear pooling computes

$$B = \frac{1}{HW} F F^T$$

which captures second-order feature interactions. After bilinear pooling, the resulting matrix was flattened, followed by signed square-root normalization and L_2 normalization. These steps are standard in bilinear representations because they stabilize the heavy-tailed distribution of second-order statistics before classification. The final classifier was a linear layer on top of the normalized bilinear descriptor.

The B-CNN was trained using cross-entropy loss and AdamW with differential learning rates: a lower learning rate for the pretrained feature extractor and a higher learning rate for the newly initialized classifier. This design allowed the model to preserve the useful pretrained convolutional features while adapting the bilinear head more aggressively to texture recognition.

4.8 Transformer-Based Model Design

4.8.1 ViT-Lite Baseline

As a lightweight Transformer baseline, the project implemented a ViT-lite pipeline based on a pretrained DeiT-Small model, following the broader Vision Transformer paradigm [14]. This model operated on 224×224 input and used Mixup, cosine annealing, and AdamW optimization. The rationale for including this model was to provide a Transformer comparison point that remained relatively lightweight while still representing the self-attention paradigm.

For both DTD and KTH-TIPS2-b, the ViT-lite design used a straightforward fine-tuning scheme in which the pretrained classifier was replaced with a new head matching the target number of classes. Compared with the CNN baselines, the training strategy relied more strongly on AdamW and moderate Mixup rather than selective layer freezing. This reflected the different optimization requirements of Transformer models.

4.8.2 Swin-Tiny at 224×224

The next step in the design progression was Swin-Tiny [15]. This model retained the 224×224 resolution but introduced hierarchical feature extraction and shifted-window self-attention. In the codebase, Swin-Tiny was created through timm.create_model('swin_tiny_patch4_window7_224', pretrained=True, num_classes=...). It was trained using cross-entropy loss, AdamW, cosine annealing, AutoAugment, and Mixup with alpha = 0.2.

Swin-Tiny served an important intermediate role. It allowed the study to evaluate whether a hierarchical Transformer could already improve material recognition without immediately incurring the cost of a large 384-resolution model. Thus, it helped isolate the effect of architectural change before the final performance-driven configuration was introduced.

4.8.3 Swin-Base at 384×384

The strongest configuration in the project was Swin-Base with 384×384 input [15]. This model was implemented through timm.create_model('swin_base_patch4_window12_384', pretrained=True, num_classes=...). The increase in resolution was intended to preserve more texture detail, which is particularly important for material classification where fine spatial patterns may be discriminative.

Because the larger model and higher input resolution significantly increased memory usage, the design incorporated gradient accumulation. The physical batch size was set to 16, and the accumulation step was set to 4, resulting in an effective batch size of 64. The loss in each mini-batch was divided by the accumulation factor before backpropagation, and the optimizer was stepped only after the required number of mini-batches had been accumulated. This allowed the model to benefit from a larger optimization batch without exceeding hardware limits.

The Swin-Base-384 pipeline also used AdamW with a lower learning rate of 3×10^{-5}, weight decay of 0.05, AutoAugment, Mixup with $\alpha = 0.2$, and cosine annealing scheduling. This final design reflects the mature stage of the project, where architectural capacity, higher-resolution input, and more careful optimization were combined to maximize cross-domain recognition performance.

4.9 Optimization Strategy and Checkpoint Selection

Although different model families used different architectural assumptions, the optimization logic was kept broadly consistent. All trainable models used cross-entropy as the core classification objective. CNN-based baselines primarily used Adam, while B-CNN and Transformer-based models used AdamW, which is generally better suited to decoupled weight decay and more stable Transformer fine-tuning. Cosine annealing learning-rate scheduling was employed across the major pipelines to reduce the learning rate gradually and smoothly over the course of training.

Model-specific hyperparameters were selected according to architecture size and training behaviour. The ResNet-based DTD regularized baseline used 40 epochs, while MobileNetV2, B-CNN, ViT-lite, and Swin models generally used 50 to 60 epochs. Standard batch size was 32 for 224-resolution models when memory allowed, while 384-resolution B-CNN and Swin-Base models used batch size 16. Differential learning rates were used in B-CNN to stabilize the pretrained backbone while adapting the new classifier. Lower fine-tuning rates were used for the largest Transformer models to avoid unstable optimization.

Across all trainable pipelines, the best checkpoint was selected according to validation accuracy. After training, the best-performing model state was reloaded and used for final evaluation. This design ensured that the final test or target-domain results were not simply taken from the last epoch, which is especially important in small-sample settings where validation performance may fluctuate.

4.10 Unified Evaluation Framework

To ensure comparability across model families, the project adopted a unified evaluation framework. The primary metric was classification accuracy, since it provides a clear overall measure of correctness. However, because texture recognition often involves visually similar categories, accuracy alone was not considered sufficient. Therefore, precision, recall, and F1-score were also computed from the classification reports, and confusion matrices were generated for detailed error analysis.

The confusion matrix was particularly important in this study because many failures were expected to arise from structured category confusion rather than random mistakes. For example, on KTH-TIPS2-b, materials such as cotton and wool are visually similar and therefore useful for diagnosing whether a model has learned genuinely discriminative texture representations. Likewise, perfect or near-perfect class-level performance on certain categories can reveal where the learned feature space is especially well separated.

The evaluation code was designed to support these analyses directly. After the best checkpoint was selected, the model predictions and ground-truth labels were collected across the target dataset, and the classification report and confusion matrix were generated using scikit-learn tools. The same evaluation principle was used for CNN, hybrid, B-CNN, and Transformer models, ensuring that later comparisons in Chapter 5 could be made on a consistent basis.

In addition to these recognition metrics, the broader project also considered real-time feasibility and robustness to noise as important practical dimensions. While the detailed comparative discussion of these aspects is reserved for Chapter 5, the design of the experiments already reflected this concern through the inclusion of both lightweight and high-capacity models, as well as through the comparison of 224-resolution and 384-resolution variants.

4.11 Relation to the Joint Project

Since this work forms part of a joint final year project, the design of the experiments also had to support later cross-comparison with the collaborating part. For this reason, the present thesis did not isolate recognition accuracy from deployment considerations. Instead, the model set was deliberately chosen to span different points on the capacity–efficiency spectrum: lightweight MobileNetV2, conventional ResNet baselines, hybrid CNN–SVM, texture-specialized B-CNN, ViT-lite, and Swin Transformer variants.

This structure enables the present thesis to provide the representational and methodological backbone of the joint study. In other words, the main contribution of this chapter is not merely the training of a single best model, but the construction of an experimental system in which different modelling philosophies can be compared fairly under controlled data regimes. This in turn allows the collaborating work to be interpreted alongside the results of this thesis in a more deployment-oriented context.

4.12 Chapter Summary

This chapter has described the project design used in the present study. The overall framework was organized into two stages: a small-sample DTD stage focused on overfitting diagnosis and mitigation, and a KTH-TIPS2-b stage focused on cross-sample generalization under a strict leave-one-sample-out protocol. The chapter then detailed the preprocessing and augmentation strategy, the design of the CNN baselines, the hybrid ResNet50+SVM pipeline, the texture-specialized B-CNN, and the Transformer-based models including ViT-lite, Swin-Tiny, and Swin-Base-384. Finally, it explained the optimization settings, checkpoint selection strategy, and unified evaluation framework used throughout the project.

These design choices provide the basis for the next chapter, which presents the experimental results and analyzes how the different model families behaved under the two benchmark settings.

CHAPTER 5 Results

5.1 Introduction

This chapter presents the experimental results of the proposed study and analyzes how different model families behaved under two distinct evaluation regimes. The first part focuses on the Describable Textures Dataset (DTD), where the main objective was to diagnose and address the small-sample overfitting problem. The second part focuses on KTH-TIPS2-b under a strict leave-one-sample-out setting, where the emphasis shifts from optimization stability to cross-sample generalization. In addition to quantitative recognition results, this chapter also incorporates a deployment-oriented CPU benchmark in order to support the "real-time" aspect of the thesis more directly.

In line with the design described in Chapter 4, the results are discussed from three perspectives. First, the overall quantitative performance of each model is compared using precision, recall, F1-score, and, where explicitly available, reported accuracy. Second, confusion matrices are analyzed to identify structured error patterns, especially for visually similar categories. Third, the findings are interpreted in relation to deployment considerations, including computational efficiency, practical real-time feasibility, and joint recommendations derived from both branches of the final year project.

The collaborating branch of the joint project later provided a DTD-centred comparative study rather than a full latency-oriented deployment report. Therefore, the joint discussion in Sections 5.4 and 5.5 focuses on complementary evidence from the two branches and derives a deployment-oriented recommendation without overstating unavailable measurements.

5.2 Results on the DTD Dataset

5.2.1 Small-Sample Overfitting in End-to-End CNN Training

The first stage of the project used DTD to investigate how different models behaved in a low-data regime. The earliest finding was that standard end-to-end deep fine-tuning was highly vulnerable to overfitting. In the initial ResNet-50 experiment, the training accuracy rose to 82%, whereas the validation accuracy remained around 65%, and the final test accuracy was only 63.94%. This result indicates that the model was fitting the training distribution much more effectively than it was learning transferable texture representations. Since DTD contains only a limited number of training samples per class, this behaviour is consistent with memorization rather than robust generalization.

Even after moving to lighter backbones and stronger regularization, the underlying difficulty remained. The DTD setting therefore served not merely as a standard classification benchmark, but as a diagnostic testbed for small-sample instability. The central question in this stage was not only which model performed best, but also which modelling strategy could remain stable under severely constrained supervision.

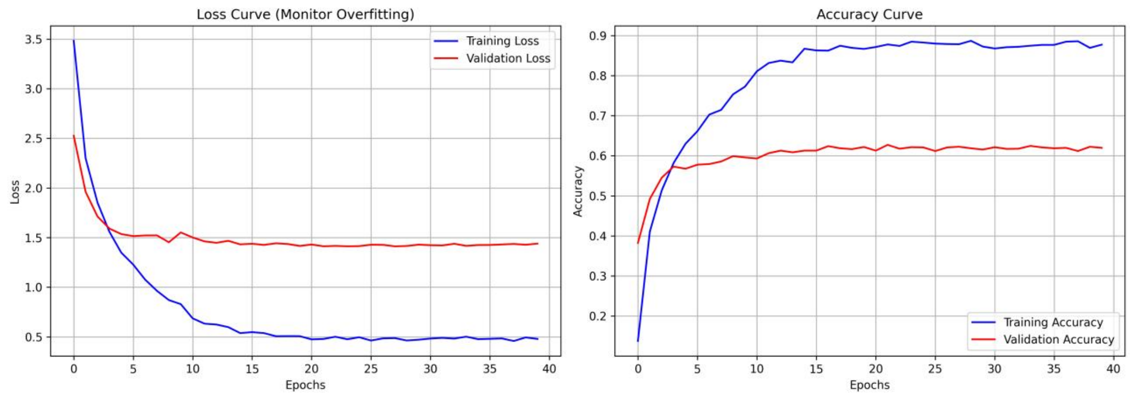

Figure 5.1 Training and validation curves of the initial DTD baselines

5.2.2 Effect of Regularization and Lightweight Backbones

To reduce overfitting, the study next introduced lighter CNNs and stronger regularization. The ResNet18 baseline achieved a macro precision of 0.6480, macro recall of 0.6426, and macro F1-score of 0.6388. MobileNetV2, combined with stronger augmentation and Mixup, achieved a macro precision of 0.6514, macro recall of 0.6415, and macro F1-score of 0.6397. Although MobileNetV2 showed more stable training behaviour than the earlier overfitted ResNet-50, its final improvement remained limited.

This outcome suggests that overfitting and underfitting formed a trade-off in the DTD regime. A heavy backbone could memorize the training set, whereas a lightweight backbone with aggressive regularization could become too constrained to extract sufficiently discriminative class boundaries. Therefore, the issue was not simply the absence of regularization, but the instability of end-to-end optimization itself when only limited labeled texture data were available.

The confusion matrices of these baselines support this interpretation. Both models showed a dispersed pattern of off-diagonal errors across multiple texture categories, indicating that the resulting decision boundaries were still not sufficiently sharp. Although the diagonal remained meaningful, the predictions were less concentrated than in the stronger later-stage models.

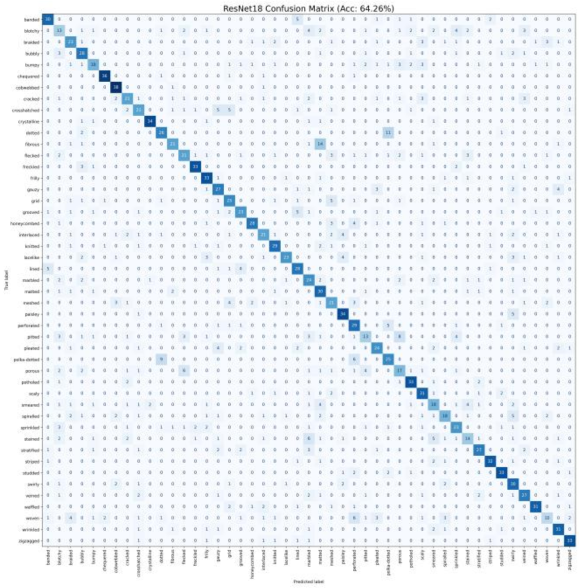

Figure 5.2 Confusion matrix of ResNet18 on DTD

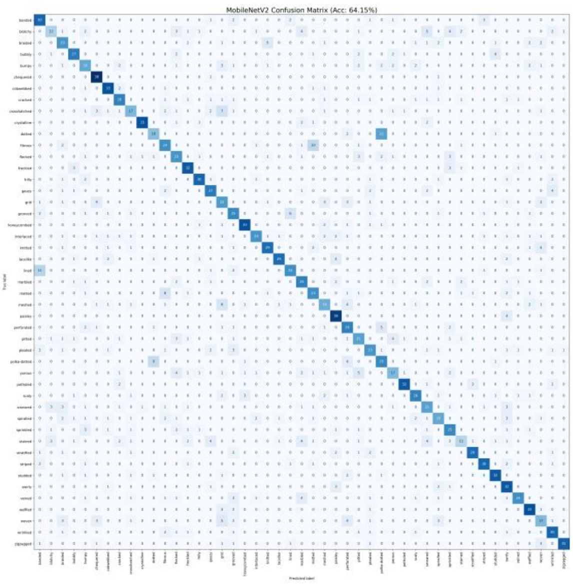

Figure 5.3 Confusion matrix of MobileNetV2 on DTD

5.2.3 Success of the Hybrid Strategy

The next design step was to decouple representation learning from final classification. Instead of continuing to fine-tune a full neural classifier on DTD, the study adopted a hybrid pipeline in which a pretrained ResNet-50 served as a frozen feature extractor and a linear SVM was trained on the extracted features. This design improved the DTD macro precision to 0.7092, macro recall to 0.7064, and macro F1-score to 0.7051, thereby clearly outperforming the earlier end-to-end CNN baselines.

This result is significant because it shows that the pretrained convolutional representation itself was not the problem. Rather, the difficulty lay in fitting a stable end-to-end classifier under scarce supervision. By separating the representation stage from the final decision boundary, the hybrid model retained the benefits of deep features while avoiding unstable low-data fine-tuning.

The confusion matrix of the hybrid model further confirms this interpretation. Compared with the ResNet18 and MobileNetV2 baselines, the diagonal becomes more concentrated and the off-diagonal scattering is reduced. In other words, the hybrid design improved not only aggregate metrics but also the structure of the prediction matrix itself.

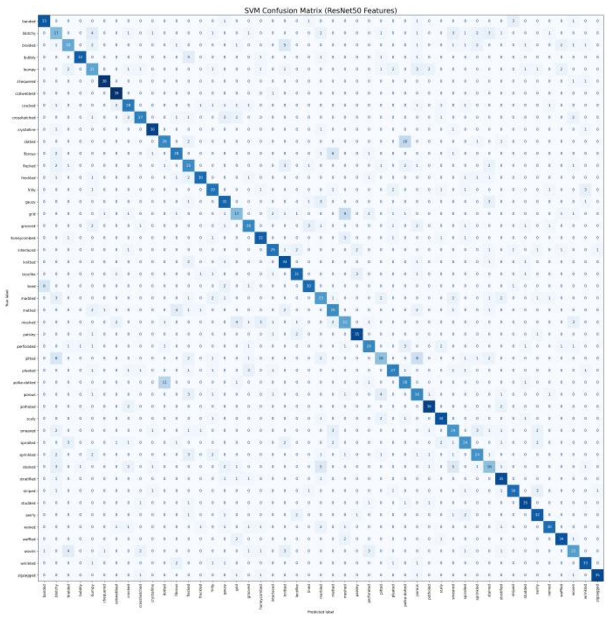

Figure 5.4 Confusion matrix of ResNet50+SVM on DTD

5.2.4 Extended DTD Comparison with Stronger Models

After establishing the usefulness of the hybrid strategy, the study also evaluated stronger representation-driven architectures on DTD. ViT-Lite achieved a macro precision of 0.7176, macro recall of 0.7190, and macro F1-score of 0.7070, representing a modest improvement over the hybrid baseline. B-CNN further improved the DTD macro F1-score to 0.7464, which indicates that second-order feature interactions can be beneficial for fine-grained texture discrimination. The strongest result in this branch was achieved by Swin Transformer, which reached a macro precision of 0.8620, macro recall of 0.8564, and macro F1-score of 0.8566.

These results show a clear architectural progression on DTD. Conventional end-to-end CNNs were limited by unstable optimization, the hybrid CNN–SVM design improved stability, B-CNN provided stronger texture-specific aggregation, and Swin Transformer offered the cleanest overall representation among all tested models in this branch.

Model	Macro Precision	Macro Recall	Macro F1-score
ResNet18	0.6480	0.6426	0.6388
MobileNetV2	0.6514	0.6415	0.6397
ResNet50 + SVM	0.7092	0.7064	0.7051
ViT-Lite	0.7176	0.7190	0.7070
B-CNN	0.7539	0.7491	0.7464
Swin Transformer	0.8620	0.8564	0.8566

Table 5.1 Macro-level results on DTD in the present branch

5.2.5 Confusion-Matrix Analysis on DTD

The DTD confusion matrices reveal that the challenge of this dataset lies not in random prediction noise, but in diffuse ambiguity across many fine-grained semantic texture categories. In the weaker CNN baselines, off-diagonal errors remain scattered throughout the matrix, which indicates that the learned feature space is not sharply separated. This is consistent with the semantic nature of DTD, where categories are often differentiated by subtle descriptive attributes rather than by strongly distinct material structure.

By contrast, the stronger later-stage models show a progressively cleaner diagonal. ViT-Lite reduces some of the broader ambiguity seen in the CNN baselines, B-CNN further sharpens separation by modelling higher-order feature correlations, and Swin Transformer produces the most concentrated confusion matrix among all tested models. The DTD results therefore establish the first major finding of this thesis: under severe small-sample constraints, conventional end-to-end CNN fine-tuning is often suboptimal, and stronger representation-driven strategies offer a more reliable path to improvement.

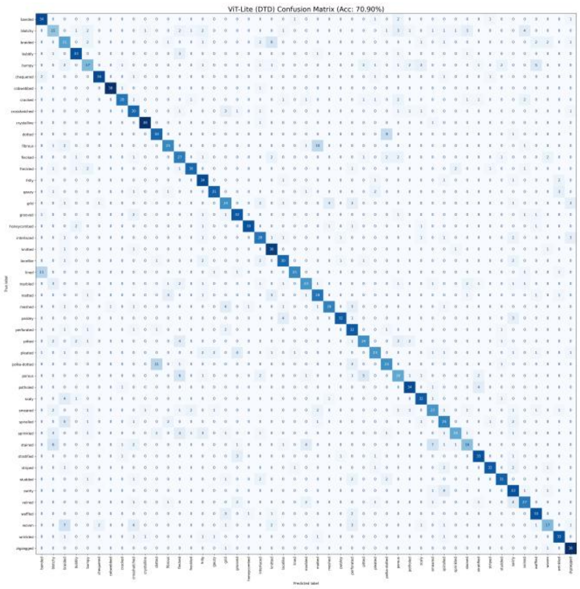

Figure 5.5 Confusion matrix of ViT-Lite on DTD

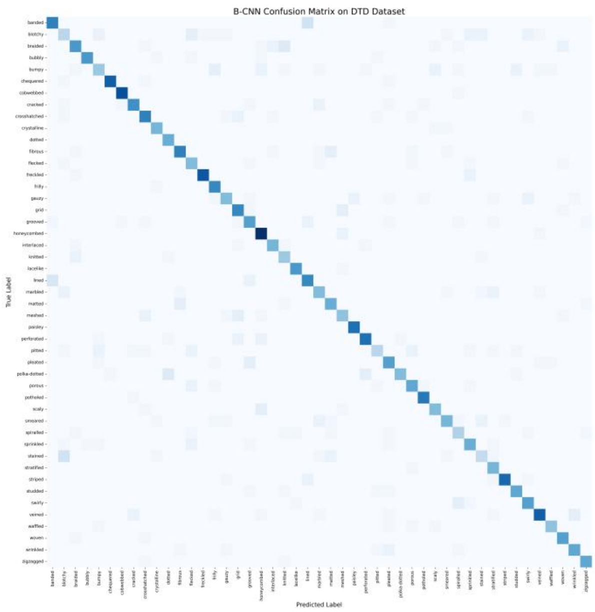

Figure 5.6 Confusion matrix of B-CNN on DTD

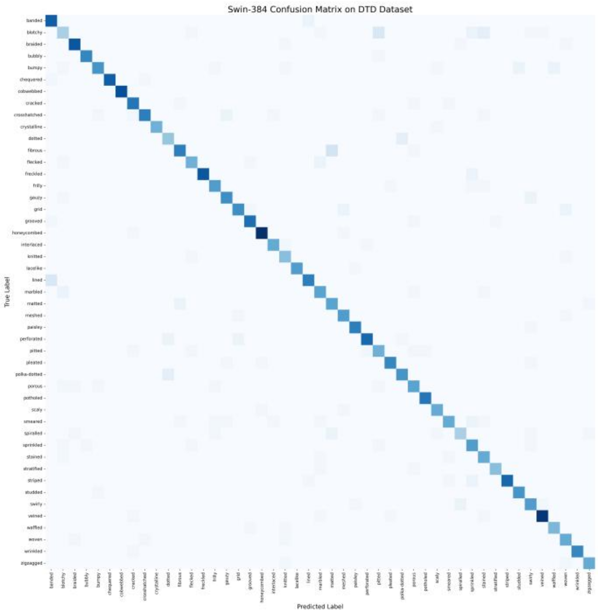

Figure 5.7 Confusion matrix of Swin Transformer on DTD

5.3 Results on KTH-TIPS2-b

5.3.1 Overall Quantitative Comparison Under Strict Cross-Sample Evaluation

The second stage of the project used KTH-TIPS2-b under a strict leave-one-sample-out protocol, where the model was trained on samples A, B, and C and tested on sample D. This made the task substantially different from the DTD setting. The primary challenge was no longer only data scarcity, but also domain shift across physically different samples of the same material.

The macro-level results are summarized in Table 5.2. The ResNet18 baseline achieved a macro precision of 0.8144, macro recall of 0.7955, and macro F1-score of 0.7964. MobileNetV2 obtained a macro precision of 0.7715, macro recall of 0.7668, and macro F1-score of 0.7541, showing that lightweight efficiency alone was insufficient for this stricter material-recognition setting. The hybrid ResNet50+SVM pipeline improved the macro F1-score to 0.8125, again demonstrating the value of decoupled classification.

ViT-Lite further improved the macro F1-score to 0.8357, while B-CNN reached a reported accuracy of 86.03% and a macro F1-score of 0.8545. Swin-Tiny at 224 resolution achieved a reported accuracy of 87.46%, indicating that hierarchical attention already provided a strong improvement over the earlier baselines. The best result was obtained by Swin-Base at 384×384 resolution, which achieved a reported accuracy of 90.57% together with a macro precision of 0.9027, macro recall of 0.9057, and macro F1-score of 0.9018.

Model	Reported Accuracy	Macro Precision	Macro Recall	Macro F1-score
ResNet18	--	0.8144	0.7955	0.7964
MobileNetV2	--	0.7715	0.7668	0.7541
ResNet50 + SVM	--	0.8321	0.8199	0.8125
ViT-Lite	--	0.8369	0.8418	0.8357
B-CNN	86.03%	0.8584	0.8603	0.8545
Swin-Tiny (224)	87.46%	--	--	--
Swin-Base (384)	90.57%	0.9027	0.9057	0.9018

Table 5.2 Results on KTH-TIPS2-b in the present branch

These results show a clearer performance hierarchy than on DTD. Under cross-sample material recognition, stronger architectures consistently provided larger gains, and the best-performing model was a large hierarchical Transformer with higher input resolution.

5.3.2 Confusion-Matrix Analysis of KTH-TIPS2-b

The KTH-TIPS2-b confusion matrices show a much more structured pattern than the DTD matrices. Here, the main source of error is concentrated in a small number of visually similar material classes rather than being broadly distributed across the full label space.

The ResNet18 baseline classified several categories relatively well, including aluminium foil, lettuce leaf, and white bread, but struggled more clearly with cotton, linen, and wool. MobileNetV2 showed even weaker performance on the hardest classes, suggesting that its compact architecture sacrifices too much discriminative capacity in this more demanding setting.

The hybrid ResNet50+SVM model improved the overall balance of the confusion matrix and produced cleaner predictions for several categories. However, visually ambiguous material pairs still remained challenging, which suggests that the limitation had shifted from optimization instability to feature-level similarity.

ViT-Lite further reduced broad confusion and improved the recognition of several easy categories. B-CNN produced a stronger diagonal than the earlier baselines and confirmed the value of second-order feature interactions for texture analysis. Nevertheless, cotton and wool remained difficult even for B-CNN. The strongest confusion structures were obtained by Swin-Tiny and especially Swin-Base-384. The Swin-Base matrix showed near-perfect recognition for multiple categories, while the remaining residual errors were concentrated mainly among cotton, linen, and wool.

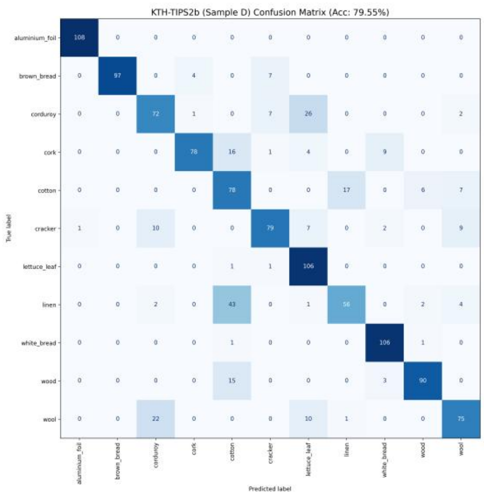

Figure 5.8 Confusion matrix of ResNet18 on KTH-TIPS2-b

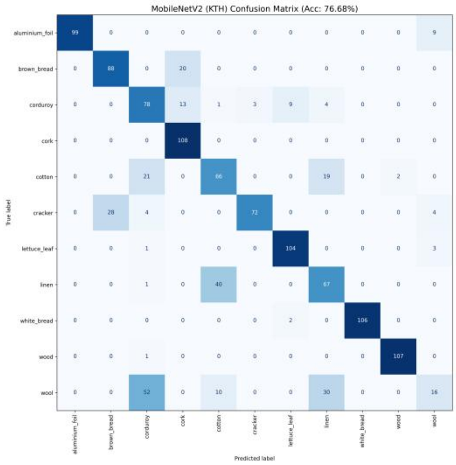

Figure 5.9 Confusion matrix of MobileNetV2 on KTH-TIPS2-b

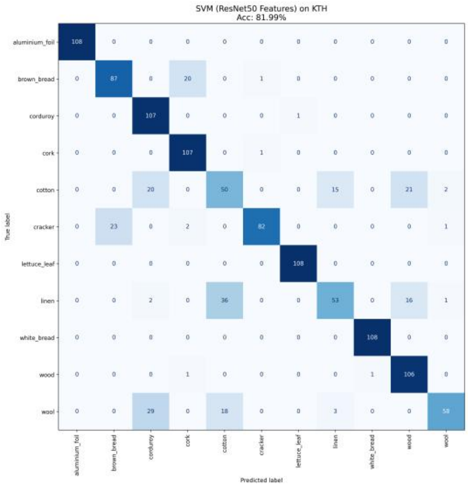

Figure 5.10 Confusion matrix of ResNet50+SVM on KTH-TIPS2-b

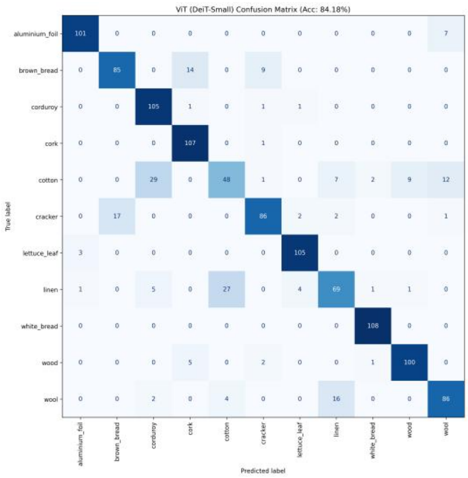

Figure 5.11 Confusion matrix of ViT-Lite on KTH-TIPS2-b

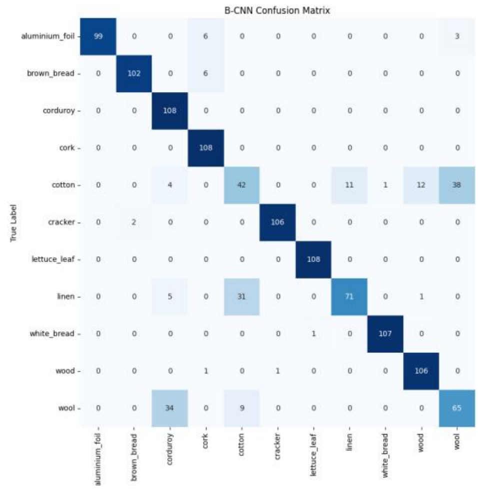

Figure 5.12 Confusion matrix of B-CNN on KTH-TIPS2-b

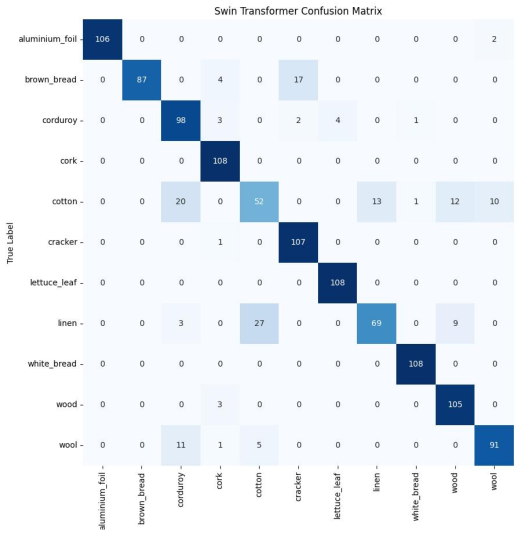

Figure 5.13 Confusion matrix of Swin-Tiny on KTH-TIPS2-b

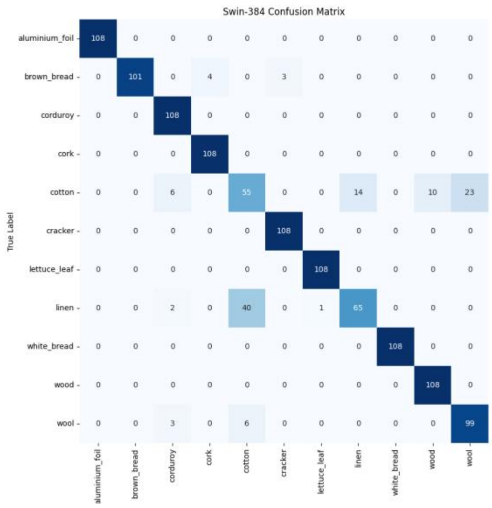

Figure 5.14 Confusion matrix of Swin-Base-384 on KTH-TIPS2-b

5.3.3 Why Swin-Base-384 Performed Best

The superior performance of Swin-Base-384 can be attributed to the combination of three factors. First, the hierarchical Transformer architecture provided stronger multi-scale modelling than the earlier CNN baselines and the lighter ViT-Lite model. This is particularly important in texture recognition, where local micro-patterns and broader spatial arrangement both contribute to class discrimination. Second, the 384×384 input resolution preserved more fine-grained texture detail than the 224×224 setting, which likely improved material separation in ambiguous classes. Third, the final training setup, including AutoAugment, Mixup, AdamW, and gradient accumulation, allowed the large model to be optimized more effectively.

Thus, Swin-Base-384 did not simply improve already easy categories. Its main contribution was to reduce structured confusion among difficult material classes, which is precisely what was needed to surpass the earlier baselines and reach 90.57% accuracy.

5.4 Discussion

5.4.1 Main Findings Across the Two Datasets

Taken together, the results from the two stages indicate that model preference is strongly dependent on the data regime. On DTD, the dominant challenge was small-sample instability. In this setting, the main contribution of the present branch was to show that a hybrid CNN–SVM strategy could outperform conventional end-to-end CNN fine-tuning once the latter became either overfitted or underfitted. In particular, the hybrid ResNet50+SVM pipeline improved the DTD macro F1-score to 0.7051, surpassing the earlier end-to-end CNN baselines, while later representation-strong models such as B-CNN and Swin achieved even stronger results.

On KTH-TIPS2-b, the dominant challenge shifted from small-sample instability to cross-sample domain shift. Under this stricter setting, architectural expressiveness became more important. B-CNN showed that texture-specific aggregation was useful, and Swin-based models showed that hierarchical self-attention was even more effective when higher-resolution input and stronger optimization were combined. Therefore, the results do not support a single universally best method. Instead, they show that model choice should be matched to the dominant difficulty of the dataset.

This regime-dependent interpretation also aligns well with the overall motivation of the thesis. In low-data settings, classifier stability and representation reuse are especially important, whereas in more challenging cross-sample recognition settings, model capacity and hierarchical context modelling become the main factors.

5.4.2 Real-Time Feasibility and Robustness to Variation

In addition to recognition quality, deployment-oriented feasibility was evaluated through a CPU-based inference benchmark, since CPU inference with batch size 1 is more representative of practical edge-side or resource-constrained scenarios than throughput-oriented GPU testing. In this study, pure inference time was measured without disk I/O over 100 runs. The benchmark included three representative model types: MobileNetV2 as the lightweight baseline, the hybrid CNN–SVM pipeline as the small-sample-stable alternative, and the tested Swin configuration as the highest-performing model family.

The benchmark results are summarized in Table 5.3. MobileNetV2 achieved the fastest inference speed, with an average latency of 49.79 ms per image and a throughput of 20.08 FPS. The hybrid CNN–SVM pipeline required 87.96 ms per image, corresponding to 11.37 FPS. Although slower than MobileNetV2, it remained substantially faster than the tested Swin configuration and still operated within a practically meaningful range for lower-rate inspection scenarios. By contrast, Swin Transformer showed the highest computational cost, with an average latency of 681.05 ms per image and only 1.47 FPS on CPU. This confirms that the gain in recognition accuracy is accompanied by a clear deployment penalty when the hardware environment is limited.

Model	Representative Role	Avg Latency (ms/img)	FPS	Deployment Interpretation
MobileNetV2	Lightweight baseline	49.79	20.08	Best choice when CPU-side real-time responsiveness is prioritized
ResNet50 + SVM	Hybrid small-sample solution	87.96	11.37	Good compromise between stability and deployment practicality
Swin Transformer	Highest-performance model	681.05	1.47	Best recognition performance, but least suitable for CPU-side real-time deployment

Table 5.3 Deployment-oriented comparison of representative models

These results make the performance–efficiency trade-off much more explicit. MobileNetV2 is the most suitable option when real-time responsiveness and lightweight deployment are prioritized. The hybrid CNN–SVM model occupies an intermediate position: it offers clearly better stability in limited-data settings than ordinary CNN fine-tuning while maintaining moderate inference speed. Swin Transformer remains the strongest option in terms of recognition performance, but its CPU-side latency suggests that it is better suited to scenarios where accuracy is the dominant objective and stronger hardware support is available.

With respect to robustness, the currently available evidence remains primarily qualitative. Neither the present branch nor the collaborator's report includes a dedicated synthetic-noise benchmark with numerical robustness scores. Therefore, no formal robustness ranking is claimed here. However, the confusion-matrix evidence from both branches suggests that stronger models reduce structured class ambiguity more effectively. In particular, Swin-based models produce the cleanest diagonals on both

DTD and KTH-TIPS2-b, while MobileNetV2 maintains a reasonable balance between discriminative power and computational efficiency. This suggests that, in practice, deployment decisions should not be based on speed alone, but on the joint consideration of recognition performance, confusion behaviour, and computational budget.

5.5 Comparison with the Collaborating Part of the Joint Project

The collaborating branch of the joint project focused primarily on a DTD-centred comparison across major model families and showed a consistent progression in performance from conventional CNNs to Transformer-based architectures. Its reported DTD results are summarized in Table 5.4. The collaborator reported macro F1-scores of 0.6304 for ResNet18, 0.6835 for ResNet50, 0.7109 for MobileNetV2, 0.7395 for B-CNN, 0.7415 for ViT, and 0.8502 for Swin Transformer. The accompanying confusion-matrix analysis also indicated that model evolution was associated with progressively cleaner diagonals and fewer structured category confusions.

Model	Precision	Recall	F1-score
ResNet18	0.6355	0.6346	0.6304
ResNet50	0.6866	0.6781	0.6835
MobileNetV2	0.7015	0.7142	0.7109
B-CNN	0.7335	0.7319	0.7395
ViT	0.7464	0.7382	0.7415
Swin Transformer	0.8525	0.8589	0.8502

Table 5.4 DTD results reported in the collaborating branch

The present thesis complements that DTD-centred comparison in two ways. First, it provides a detailed analysis of the small-sample optimization problem and demonstrates that a hybrid CNN–SVM strategy can outperform conventional end-to-end CNN fine-tuning when the data regime is highly constrained. Second, it extends the study to KTH-TIPS2-b under a strict leave-one-sample-out protocol, where Swin-Base at 384×384 achieved the strongest overall recognition result, reaching 90.57% accuracy and 0.9018 macro F1-score.

When the two branches are viewed together, their findings are complementary rather than redundant. The collaborator's branch is valuable in showing, within a single

DTD comparison, that the model ranking from conventional CNNs to Swin is structurally consistent and that MobileNetV2 deserves attention as an efficiency-oriented baseline. The present thesis adds that architectural superiority depends on the nature of the task: hybrid learning is especially useful when data scarcity causes unstable convergence, whereas large hierarchical Transformers become most beneficial when the problem shifts toward stronger domain variation and fine-grained material ambiguity.

The newly added CPU-side benchmark further strengthens this joint interpretation. Under batch size 1 and excluding disk I/O, MobileNetV2 achieved the fastest inference speed at 49.79 ms per image (20.08 FPS), the hybrid CNN–SVM pipeline achieved 87.96 ms per image (11.37 FPS), and the tested Swin configuration required 681.05 ms per image (1.47 FPS). These measurements make clear that the joint project does not point to a single universally optimal model. Instead, it supports a tiered recommendation. If maximum recognition accuracy is the dominant objective, a Swin-based architecture should be preferred. If CPU-side real-time responsiveness is more important, MobileNetV2 is the most practical candidate among the tested models. If the available labeled data are limited and stable convergence is a concern, the hybrid CNN–SVM strategy provides a useful middle ground between recognition performance and deployment cost.

Therefore, the final recommendation of the joint project is scenario-dependent rather than absolute. Swin-based models define the upper bound of recognition performance, MobileNetV2 remains the strongest lightweight option among the tested models, and the hybrid CNN–SVM approach is especially valuable in small-sample conditions where conventional fine-tuning becomes unreliable. This conclusion is more meaningful for industrial deployment than reporting a single “best model,” because it directly reflects the real trade-off among accuracy, stability, and inference efficiency.

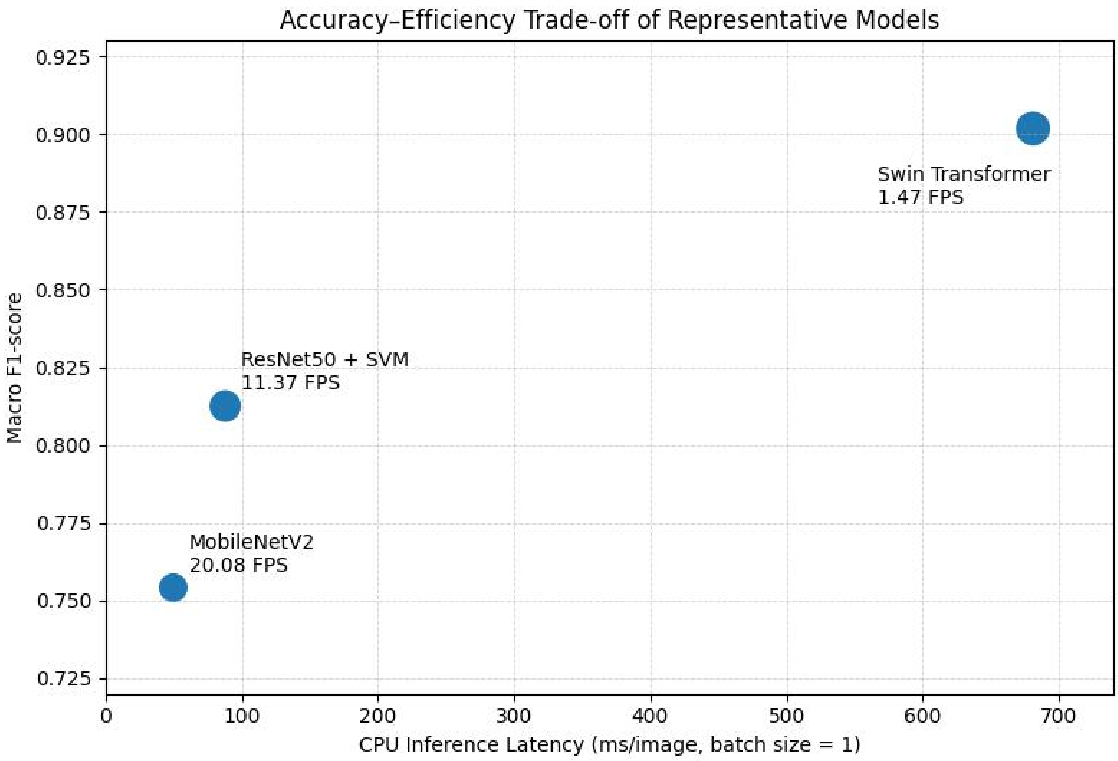

Figure 5.15 Joint accuracy－efficiency trade-off of representative models

A practical design for Figure 5.15 is a scatter plot with average CPU latency on the horizontal axis and macro F1-score on the vertical axis. MobileNetV2 should appear on the fast but lower-performing side, Swin Transformer on the high-performing but slow side, and the hybrid CNN–SVM model between the two. This figure visually summarizes the final joint recommendation of the project.

5.6 Chapter Summary

This chapter has presented the experimental results of the study across two different texture-recognition regimes. On DTD, the experiments showed that conventional end-to-end CNN fine-tuning was strongly affected by the small-sample problem, and that the hybrid ResNet50+SVM strategy provided a more stable and effective alternative. On KTH-TIPS2-b, the results showed a clear progression from CNN baselines to texture-specialized deep models and finally to hierarchical Transformers, with Swin-Base-384 achieving the best overall result of 90.57% accuracy and 0.9018 macro F1-score under a strict cross-sample protocol.

The confusion-matrix analysis further showed that the main challenge on KTH-TIPS2-b was concentrated in a few visually similar classes, especially cotton, linen, and wool, and that the strongest models improved performance primarily by reducing this structured confusion. In addition, the comparison with the collaborating branch showed that the conclusions of the joint project are consistent at a broader level: Swin-based models provide the strongest recognition performance, MobileNetV2 remains relevant as a lightweight candidate, and hybrid CNN–SVM design is particularly useful under severe small-sample constraints.

The added CPU-side benchmark further showed that the final model recommendation depends not only on recognition quality but also on deployment constraints: MobileNetV2 was the fastest tested model at 20.08 FPS, the hybrid CNN–SVM pipeline achieved a moderate 11.37 FPS, and the tested Swin configuration, although most accurate, ran at only 1.47 FPS on CPU. These findings provide the basis for the next chapter, which discusses future work and the remaining challenges for lightweight deployment and edge-side implementation.

CHAPTER 6 Future Work

6.1 Introduction

The experimental results in Chapter 5 demonstrate that different model families have different advantages under different texture-recognition conditions. Hybrid CNN–SVM models are effective under small-sample constraints, while Swin-based Transformers achieve the strongest recognition performance under more challenging cross-sample settings. However, the deployment-oriented benchmark also shows that the highest-performing model is not necessarily the most practical model for real-time industrial use. In particular, the tested Swin configuration achieved the best recognition quality but only reached 1.47 FPS on CPU, whereas MobileNetV2 achieved 20.08 FPS but with lower recognition performance.

Therefore, future work should focus not only on further improving accuracy, but also on narrowing the gap between high-performance recognition and practical deployment. This chapter discusses several directions for future improvement, including model lightweighting, edge-hardware implementation, robustness evaluation, industrial data expansion, and interpretability.

6.2 Model Lightweighting and Compression

The first direction for future work is model lightweighting. The results of this study show that Swin-based models provide the strongest recognition performance, but their computational cost is considerably higher than that of lightweight CNNs. This creates a practical limitation for real-time inspection systems, especially when deployment must be performed on CPUs, embedded devices, or edge platforms with limited memory and power budgets.

Several model-compression strategies can be explored in future work. One possible approach is knowledge distillation, where a large high-performing model such as Swin-Base acts as the teacher and a smaller model such as MobileNetV2 or Swin-Tiny serves as the student. The goal would be to transfer the discriminative knowledge of the teacher model into a more efficient architecture. This could allow the student model to retain part of the accuracy advantage of Swin while maintaining faster inference speed.

Another possible direction is pruning, which removes redundant channels, layers, or attention heads from a trained model. Since texture recognition often depends on repeated local patterns, it is possible that not all channels or attention heads contribute equally to classification. Structured pruning could therefore reduce model size while preserving the most useful texture-sensitive components. Quantization could also be applied to reduce numerical precision from full floating-point operations to lower-bit representations, which may improve inference speed and reduce memory consumption on supported hardware.

For this project, a particularly meaningful future experiment would be to compare MobileNetV2, Swin-Tiny, and a distilled lightweight Transformer under the same CPU benchmark used in Chapter 5. This would directly test whether lightweight models can approach Swin-level accuracy while remaining suitable for real-time deployment.

6.3 Edge-Hardware Deployment

The current deployment evaluation was conducted using CPU inference with batch size 1, which provides an initial indication of real-time feasibility. However, a complete deployment study would require testing on actual edge hardware. In industrial inspection systems, models may be deployed on devices such as NVIDIA Jetson boards, industrial PCs, mobile processors, or dedicated AI accelerators. The performance characteristics on these platforms may differ significantly from those observed on a general-purpose CPU.

Future work should therefore include hardware-specific deployment experiments. For example, the models can be exported to deployment formats such as ONNX or TensorRT and tested under realistic camera-input conditions. This would make it possible to measure not only pure inference latency, but also preprocessing time, memory usage, end-to-end frame processing speed, and power consumption. These factors are all important in real industrial systems.

A useful future benchmark would include three levels of latency measurement. The first is pure model inference, which was already tested in this thesis. The second is preprocessing plus inference, including image resizing, normalization, and tensor conversion. The third is full camera-to-decision latency, including image capture, preprocessing, model inference, post-processing, and result output. Such a benchmark would provide a more complete understanding of whether a model can truly satisfy real-time inspection requirements.

6.4 Robustness to Noise and Image Degradation

Although this thesis discusses robustness through confusion-matrix patterns and model behaviour, a dedicated quantitative robustness experiment remains an important direction for future work. In real industrial environments, images may be affected by sensor noise, motion blur, uneven illumination, compression artefacts, dust, scratches, or partial occlusion. A model that performs well on clean benchmark images may still become unreliable when image quality decreases.

Future work should therefore introduce controlled image degradation tests. Several types of noise can be added to the test images, including Gaussian noise, salt-and-pepper noise, motion blur, defocus blur, brightness shift, contrast variation, and JPEG compression. The performance degradation of each model can then be measured under increasing severity levels. This would allow a more precise comparison of robustness between MobileNetV2, CNN–SVM, B-CNN, and Swin-based models.

Such an experiment would be especially useful for understanding whether the strongest model in clean-image classification is also the most robust model under degraded imaging conditions. It is possible that larger models with stronger contextual reasoning may tolerate certain types of noise better, while lightweight models may be more sensitive to local corruption. Conversely, some high-capacity models may over-rely on fine details and degrade more quickly under blur or compression. A systematic robustness benchmark would help clarify these trade-offs.

6.5 Expansion to Real Industrial Texture and Defect Data

Another important direction is to evaluate the proposed methods on real industrial texture and defect datasets. DTD and KTH-TIPS2-b are valuable benchmarks, but they do not fully represent the complexity of semiconductor or industrial surface inspection. Real production images may contain subtle defects, rare failure modes, imbalanced class distributions, and non-standard imaging conditions. Therefore, future work should include domain-specific data collected from actual inspection scenarios.

In a real industrial dataset, the classification task may also differ from standard texture recognition. Instead of assigning each image to a general material category, the system may need to distinguish normal and defective surfaces, classify defect types, or detect small anomalous regions within a larger image. This may require extending the current image-level classification framework to patch-level classification, weakly supervised localization, or full defect detection and segmentation.

A practical future extension would be to collect a small custom dataset of industrial surface images and use the current models as baselines. The hybrid CNN–SVM strategy could be tested as a low-data baseline, while Swin-based models could be evaluated as high-performance alternatives. This would help determine whether the conclusions drawn from DTD and KTH-TIPS2-b transfer to real inspection tasks.

6.6 Semi-Supervised and Self-Supervised Learning

Data scarcity is one of the major challenges identified in this thesis. The DTD experiments showed that end-to-end CNN fine-tuning can overfit when labelled data are limited, while the hybrid approach can partially address this issue. However, a more general solution would be to reduce the dependence on manually labelled data.

Future work could explore semi-supervised and self-supervised learning for texture recognition. In semi-supervised learning, the model learns from both labelled and unlabelled images. This is useful in industrial settings because unlabelled images are often easier to collect than manually annotated defect or texture labels. Methods such as pseudo-labelling or consistency regularization could help improve generalization without requiring a large labelled dataset.

Self-supervised learning is another promising direction. A model can first learn visual representations from unlabelled industrial images through pretext tasks such as contrastive learning, masked image modelling, or image reconstruction. After this pretraining stage, the learned representation can be fine-tuned or combined with a classifier for the target texture-recognition task. This approach may be especially useful when domain-specific industrial images are available but labelled examples are scarce.

For this project, self-supervised pretraining could be particularly valuable because texture recognition depends heavily on local structure and repeated patterns. A representation learned directly from industrial texture images may be better aligned with the target domain than a backbone pretrained only on general object-recognition data.

6.7 Interpretability and Failure Analysis

Future work should also improve the interpretability of texture recognition models. In industrial inspection, it is often not enough for a model to provide a class label; engineers may also need to understand why the model made a particular prediction. This is especially important when the model is used to support quality-control decisions.

Visualization methods such as Grad-CAM, attention maps, or feature attribution can be used to analyze which regions of the image contribute most to the prediction. For CNNs, class activation maps can reveal whether the model focuses on meaningful texture regions or irrelevant background areas. For Transformer-based models, attention visualization may help show whether the model uses local texture details, global context, or both. Such analysis would be especially useful for difficult classes such as cotton, linen, and wool in KTH-TIPS2-b, where even the strongest model still showed residual confusion.

Interpretability analysis can also support model refinement. If visualization shows that the model relies on unstable background cues, stronger cropping or segmentation preprocessing may be needed. If the model focuses only on local patterns but ignores broader material structure, architectures with better context modelling may be preferable. Therefore, interpretability is not only useful for explanation, but also for improving future model design.

6.8 Future Development of the Hybrid Strategy

Although the hybrid CNN–SVM pipeline performed well under small-sample conditions, it can be further developed. The current version uses ResNet-50 as a frozen feature extractor and a linear SVM as the classifier. Future work could examine whether other feature extractors, such as EfficientNet, ConvNeXt, or lightweight Swin variants, provide better feature spaces for SVM classification.

In addition, different classifiers could be compared on top of the same deep features. For example, linear SVM, kernel SVM, logistic regression, random forest, and gradient boosting classifiers may behave differently depending on feature dimensionality and sample size. This would help determine whether the observed improvement comes mainly from the frozen deep representation, the margin-based SVM classifier, or the interaction between the two.

Another possible extension is to use multi-layer feature fusion. Instead of extracting only the final pooled CNN feature, features from multiple network stages could be combined to capture both low-level texture patterns and higher-level semantic information. This may be especially useful for texture recognition because lower layers often retain fine local details that may be weakened in deeper layers.

6.9 Chapter Summary

This chapter has discussed possible directions for future work based on the findings of the present thesis. The most immediate direction is to improve deployment feasibility through model lightweighting, compression, and edge-hardware testing. Since the CPU benchmark showed a clear gap between MobileNetV2 and Swin Transformer, future research should investigate whether distillation, pruning, quantization, or lightweight Transformer design can reduce this gap.

The chapter also identified the need for dedicated robustness evaluation, especially under noise, blur, illumination variation, and compression. In addition, future work should extend the current benchmark-based study to real industrial texture and defect datasets, where the task may involve not only classification but also anomaly detection, localization, or segmentation. Finally, semi-supervised learning, self-supervised pretraining, interpretability analysis, and further development of the hybrid CNN–SVM strategy were proposed as promising directions.

Overall, future work should aim to bridge the gap between high recognition performance and practical industrial deployment. The goal is not only to build a more accurate model, but to develop a texture recognition system that is efficient, robust, interpretable, and suitable for real camera-based inspection environments.

CHAPTER 7 Conclusions

7.1 Summary of the Study

This thesis investigated real-time texture recognition using machine learning and deep learning methods, with a particular focus on the progression from conventional convolutional neural networks to hybrid CNN–SVM pipelines and vision Transformer architectures. The study was motivated by camera-based industrial inspection scenarios, where texture recognition systems must identify subtle surface patterns accurately while also considering practical constraints such as inference speed, robustness, and deployability.

The project was conducted as part of a joint final year project. Within this broader collaboration, the present thesis focused on the algorithmic and comparative analysis of hybrid models, texture-specialized deep architectures, and Transformer-based recognition. The experimental work was organized into two main stages. The first stage used the Describable Textures Dataset (DTD) to study the small-sample problem and the limitations of end-to-end fine-tuning. The second stage used KTH-TIPS2-b under a strict leave-one-sample-out protocol to evaluate cross-sample generalization under more challenging material-recognition conditions.

Across these two stages, the thesis compared several model families, including ResNet-based CNNs, MobileNetV2, ResNet50+SVM, ViT-Lite, B-CNN, Swin-Tiny, and Swin-Base-384. The models were evaluated using precision, recall, F1-score, reported accuracy where available, confusion-matrix analysis, and CPU-side inference benchmarking. This allowed the study to move beyond a simple accuracy comparison and instead examine how different architectures behave under different data and deployment conditions.

7.2 Main Findings

The first major finding is that conventional end-to-end CNN fine-tuning is not always suitable for small-sample texture recognition. On DTD, the initial ResNet-50 experiment suffered from severe overfitting, with training accuracy reaching 82% while validation accuracy remained around 65%. Attempts to reduce this issue through lighter backbones and stronger regularization provided limited improvement. MobileNetV2 with Mixup reduced overfitting, but its performance remained around the same level as the ResNet18 baseline, indicating that the model had shifted toward underfitting rather than solving the underlying representation problem.

The second major finding is that the hybrid CNN–SVM strategy is an effective solution under limited-data conditions. By using a frozen ResNet-50 as a deep feature extractor and a linear SVM as the classifier, the model achieved a macro F1-score of 0.7051 on DTD, outperforming the earlier end-to-end CNN baselines. This result shows that the pretrained CNN features remained useful, but that stable classifier fitting was difficult when the entire network was trained end to end on limited data. Therefore, the hybrid strategy provided a practical compromise between deep representation learning and classical machine learning stability.

The third major finding is that stronger representation mechanisms become increasingly important under stricter cross-domain settings. On KTH-TIPS2-b, where the model was trained on samples A, B, and C and tested on sample D, the task required generalization to an unseen physical sample. Under this protocol, MobileNetV2 achieved a macro F1-score of 0.7541, ResNet18 achieved 0.7964, ResNet50+SVM achieved 0.8125, ViT-Lite achieved 0.8357, and B-CNN achieved 0.8545. The best result was obtained by Swin-Base-384, which achieved 90.57% accuracy and 0.9018 macro F1-score. This confirms that hierarchical Transformer architectures with high-resolution input are particularly effective for fine-grained material recognition.

The fourth major finding is that confusion-matrix analysis provides important information beyond aggregate metrics. On DTD, weaker models showed diffuse off-diagonal errors across many semantic texture categories, while stronger models produced increasingly concentrated diagonals. On KTH-TIPS2-b, the errors were more structured and concentrated among visually similar material categories, especially cotton, linen, and wool. Even the strongest Swin-Base-384 model retained some confusion among these classes, suggesting that their visual overlap represents a genuine feature-level challenge rather than a simple training failure.

The fifth major finding is that recognition accuracy and deployment efficiency are not aligned. The CPU benchmark showed that MobileNetV2 achieved the fastest inference speed at 49.79 ms per image, equivalent to 20.08 FPS. The hybrid CNN–SVM model achieved 87.96 ms per image, equivalent to 11.37 FPS. In contrast, the tested Swin configuration required 681.05 ms per image, equivalent to only 1.47 FPS on CPU. This result demonstrates that the highest-performing model is not necessarily the most suitable model for real-time CPU-side deployment.

7.3 Conclusions from the Joint Project

The joint project supports a scenario-dependent conclusion rather than a single universal model recommendation. The collaborating branch showed that, on DTD, performance increased from conventional CNNs to MobileNetV2, B-CNN, ViT, and finally Swin Transformer. This is consistent with the present thesis, where stronger representation mechanisms also produced cleaner confusion matrices and stronger classification results. At the same time, this thesis adds that model choice must be interpreted in relation to the data regime and deployment objective.

If the primary goal is maximum recognition accuracy and sufficient computational resources are available, Swin-based models are the most suitable choice among the models evaluated. They consistently achieved the strongest recognition performance and the cleanest confusion structures, especially under challenging cross-sample recognition. However, their CPU inference cost makes them less suitable for lightweight real-time deployment without further optimization.

If the primary goal is efficient CPU-side or edge-side inference, MobileNetV2 is the most practical candidate among the tested models. It achieved the fastest inference speed in the benchmark and remains relevant for deployment scenarios where frame rate and computational simplicity are more important than maximum classification accuracy.

If the primary challenge is limited labelled data and unstable end-to-end training, the hybrid ResNet50+SVM strategy is a valuable middle-ground solution. It improved over conventional CNN fine-tuning in the small-sample DTD setting and achieved moderate inference speed in the CPU benchmark. This makes it especially useful when the dataset is too small for reliable full-network optimization but pretrained deep features are still informative.

Therefore, the final recommendation of the joint project can be summarized as follows: Swin Transformer is preferred for accuracy-oriented recognition, MobileNetV2 is preferred for lightweight deployment, and CNN–SVM is preferred for small-sample stability. This three-way recommendation better reflects practical industrial requirements than choosing a single “best” model.

7.4 Contributions of the Thesis

The contributions of this thesis can be summarized in four aspects.

First, the thesis provides a comparative study of texture recognition models across different methodological families, including conventional CNNs, hybrid CNN–SVM models, texture-specialized architectures, and vision Transformers. This comparison helps clarify how model behaviour changes as the recognition task moves from small-sample semantic texture classification to stricter cross-sample material recognition.

Second, the thesis demonstrates the practical value of hybrid learning. Instead of treating classical machine learning as outdated, the study shows that an SVM classifier combined with pretrained deep features can outperform end-to-end CNN fine-tuning when labelled data are limited. This finding is useful for industrial scenarios where collecting large annotated datasets may be difficult.

Third, the thesis shows the strength of hierarchical Transformer architectures for texture recognition. The Swin-Base-384 model achieved the best result of the study on KTH-TIPS2-b, reaching 90.57% accuracy and 0.9018 macro F1-score under a strict leave-one-sample-out protocol. This supports the view that combining local attention, cross-window interaction, and high-resolution input is effective for fine-grained material classification.

Fourth, the thesis adds a deployment-oriented perspective through CPU benchmarking. By comparing MobileNetV2, ResNet50+SVM, and Swin Transformer under batch size 1, the study provides direct evidence for the trade-off between recognition performance and inference speed. This makes the final recommendation more relevant to real-time industrial inspection than a purely accuracy-based comparison.

7.5 Limitations

Despite these contributions, several limitations remain.

First, the study was conducted on public benchmark datasets rather than on a large-scale real industrial inspection dataset. Although DTD and KTH-TIPS2-b are useful for evaluating texture recognition, they cannot fully represent the complexity of semiconductor or production-line surface inspection. Real industrial data may contain rare defects, class imbalance, inconsistent lighting, and more complicated background conditions.

Second, the robustness analysis was mainly based on confusion-matrix behaviour rather than a dedicated synthetic-noise benchmark. Although the results suggest that stronger models reduce structured confusion, the study does not provide a complete quantitative robustness evaluation under noise, blur, illumination variation, or compression artefacts.

Third, the deployment benchmark measured pure CPU inference time without disk I/O. This is useful as a controlled comparison, but it does not fully represent end-to-end system latency in a real camera pipeline. A complete deployment study should also measure preprocessing time, image capture latency, memory usage, and hardware-specific acceleration.

Fourth, the comparison with the collaborating branch was complementary rather than fully unified. Since the two branches did not use exactly the same experimental protocol for every model, the joint discussion should be interpreted as a combined project-level analysis rather than a strict one-to-one benchmark across all models.

7.6 Final Remarks

In conclusion, this thesis shows that real-time texture recognition should not be approached through a single fixed model choice. Instead, the appropriate method depends on the data scale, the difficulty of the recognition problem, and the deployment constraints. Hybrid CNN–SVM models are effective when labelled data are limited and end-to-end fine-tuning is unstable. Swin-based Transformers achieve the strongest recognition performance when computational resources permit. MobileNetV2 remains valuable when real-time CPU-side inference and lightweight deployment are the main priorities.

The overall findings support a balanced and practical view of texture recognition for industrial inspection. High accuracy, robustness, and real-time efficiency cannot always be maximized simultaneously. A successful system must therefore select its model according to the specific operating scenario. This conclusion is especially relevant for camera-based industrial and semiconductor surface inspection, where both recognition reliability and deployment feasibility are essential.

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